

AP EAMCET-2018-Engineering-23 Apr-Shift-2
Mathematics

1. If $g(x) = x^2 + x - 2$ and $\frac{1}{2}(g \circ f)(x) = 2x^2 - 5x + 2$ then one such function $f(x) =$
- (1) $2x - 3$
 - (2) $2x + 3$
 - (3) $2 + 2x$
 - (d) $2x^2 - 3x - 1$
2. If $f: R \rightarrow A$ defined by $f(x) = \frac{1}{x^2 + 2x + 2}, \forall x \in R$ is subjective, then $A =$
- (1) $[1, \infty]$
 - (2) $(1, \infty)$
 - (3) $[0, 1]$
 - (4) $(0, 1]$

3. If $2 \cdot (4^{2n+1}) + 3^{3n+1}$ is divisible by k , $k > 1$ for all $n \in N$, then the value of k is

(1) 19

(2) 17

(3) 11

(4) 13

4. If $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ then the incorrect option among the following is

(1) $A^3 - I = A(A - I)$

(2) $(A^3 + I) = A(A^3 - I)$

(3) $A^4 - I = A^2 + I$

(4) $A^2 + I = A(A^2 - I)$

5. If $a \neq 1, b \neq -1, c \neq -1$ and the system of equations and the system of equations $x = a(y + z), y = b(z + x), z = c(x + y)$ has a non trivial solution then,

(1) $\frac{a}{a+1} + \frac{b}{b+1} + \frac{c}{c+1} = 0$

$$(2) \frac{a}{a+1} + \frac{b}{b+1} + \frac{c}{c+1} = 1$$

$$(3) \frac{abc}{(a+1)(b+1)(c+1)} = 1$$

$$(4) \frac{a+b+c}{(a+1)(b+1)(c+1)} = 2$$

6. The rank of following matrix A is,

$$A = \begin{bmatrix} 1 & -2 & 3 & -4 \\ 2 & 9 & 4 & 5 \\ 4 & 5 & 10 & -3 \\ 1 & 11 & -1 & 9 \end{bmatrix}$$

(1) 4

(2) 3

(3) 2

(4) 1

7. If a complex number z satisfies $|z|^2 + 1 = |z^2 - 1|$ then the locus of z is

(1) a circle

(2) the real axis

(3) the imaginary axis

(4) the straight line $y = x$

8. P is a point denoting z in the Argand diagram and if $\frac{z-i}{z+1}$ is

always purely imaginary then the locus of P is

(1) The circle with centre $\left(\frac{1}{2}, \frac{1}{2}\right)$ and radius $\frac{1}{\sqrt{2}}$

(2) The circle with centre $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ and radius $\frac{1}{\sqrt{2}}$

(3) The point on the circle with centre $\left(\frac{1}{2}, \frac{1}{2}\right)$ and radius $\frac{1}{\sqrt{2}}$,
excluding the point $(1, 0)$ and $(0, 1)$

(4) The point on the circle with centre $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ and radius
 $\frac{1}{\sqrt{2}}$, excluding the origin

9. If a, b, c are non zero real number with $c \neq 1$ such that

$$a^2 + b^2 + c^2 = c \text{ and if } \alpha = \frac{a+ib}{1-c} \text{ then } a^2 + b^2 =$$

$$(1) \frac{|\alpha|^2}{(1+|\alpha|^2)^2}$$

$$(2) \frac{|\alpha|^4}{(1+|\alpha|^2)^2}$$

$$(3) \frac{|\alpha|}{(1+|\alpha|^2)}$$

$$(4) \frac{|\alpha|}{1+|\alpha|}$$

10. For $n \in \mathbb{Z}^+$, $(1 + \sin \theta + i \cos \theta)^n + (1 + \sin \theta - i \cos \theta)^n =$

$$(1) 2^{n+1} \cdot \cos^n \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \cos \left(\frac{n\pi}{4} - \frac{\theta}{2} \right)$$

$$(2) 2^{n+1} \cdot \cos^n \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \sin \left(\frac{n\pi}{4} - \frac{\theta}{2} \right)$$

$$(3) 2^{n+1} \cdot \cos^n \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \cos \left(\frac{n\pi}{4} - \frac{n\theta}{2} \right)$$

$$(4) 2^{n+1} \cdot \cos^n \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \sin \left(\frac{n\pi}{4} - \frac{n\theta}{2} \right)$$

11. If the quadratic equation formed by eliminating x from $x^2 + \alpha x + \beta = 0$ and $xy + l(x + y) + m = 0$ has same roots as to the given quadratic equation, then the set of values of β is

- (1) $\{m, \alpha l - m\}$
- (2) $\{m, l + m\}$
- (3) $\{m, \alpha l + m\}$
- (4) $\{m, l - m\}$

12. If x is real then the range of $\frac{x^2 + 2x + 1}{x^2 + 2x + 7}$ is

- (1) $[0, 1)$
- (2) $(-\infty, 0) \cup (1, \infty)$
- (3) $(0, 1)$
- (4) R

13. If $[x]$ denotes the greatest integer not exceeding x , then the value of x satisfying $[x]^2 - 7[x] + 12 \leq 0$ is

- (1) $1 \leq x < 4$
- (2) $3 \leq x < 5$

(3) $-5 < x \leq -3$

(4) $2 \leq x \leq 4$

14. If the equation whose roots are p times the roots of

$x^4 - 2ax^3 + 4bx^2 + 8ax + 16 = 0$ is a reciprocal equation then

$|p| =$

(1) 3

(2) 2

(3) $\frac{1}{2}$

(4) $\frac{1}{3}$

15. The number of three digit numbers in which 9 appears only in one place is

(1) 243

(2) 234

(3) 217

(4) 225

16. ${}^{34}C_{10} + 3({}^{34}C_9) + 3({}^{34}C_8) + {}^{34}C_7 =$

(1) ${}^{38}C_{10}$

(2) ${}^{36}C_{10}$

(3) ${}^{37}C_{10}$

(4) ${}^{35}C_{10}$

17. If the coefficient of x^{50} in the expansion of $(1+x)^{100} + 20(1+x)^{99} + 3x^2(1+x)^{95} + \dots + 101x^{100}$, is

(1) ${}^{100}C_{50}$

(2) ${}^{101}C_{50}$

(3) ${}^{102}C_{50}$

(4) ${}^{103}C_{50}$

18. If $\alpha = \frac{5}{2! \times 3} + \frac{5 \times 7}{3! \times 3^3} + \frac{5 \times 7 \times 9}{4! \times 3^3} + \dots$, then $\alpha^2 + 4\alpha =$

(1) 21

(2) 23

(3) 25

(4) 27

19. If $\frac{x^4 + 24x^2 + 28}{(x^2 + 1)^3} = \frac{A}{x^2 + 1} + \frac{B}{(x^2 + 1)^2} + \frac{C}{(x^2 + 1)^2}$ then $A + C =$

(1) 12

(2) 10

(3) 9

(4) 6

20. If $\tan \theta = \frac{\cos 25^\circ + \sin 25^\circ}{\cos 25^\circ - \sin 25^\circ}$ and θ is in the third quadrant, then

$\theta =$

(1) 200°

(2) 205°

(3) 225°

(4) 250°

21. If $\cos A = \frac{7}{25}$ and $\frac{3\pi}{2} < A < 2\pi$ then $\cos \frac{A}{4} + \cos \frac{A}{2} - \cos 2A =$

(1) $\frac{1}{\sqrt{10}} + \frac{27}{625}$

(2) $\frac{3}{\sqrt{10}} - \frac{27}{625}$

$$(3) \frac{3}{\sqrt{10}} + \frac{27}{625}$$

$$(4) \frac{1}{\sqrt{10}} - \frac{27}{625}$$

22. In ΔABC , if $\cos 3A + \cos 3B + \cos 3C + \cos 3\pi = 0$ then the least value of the sum of two of its angles is

$$(1) \frac{\pi}{6}$$

$$(2) \frac{2\pi}{3}$$

$$(3) \frac{\pi}{3}$$

$$(4) \frac{\pi}{4}$$

23. If θ is in the interval $\left(0, \frac{\pi}{2}\right)$ satisfying the equation

$$\cos 2\theta - \sec^4 \theta + \sec^2 \theta = 0 \text{ then } \sin^2 \theta =$$

$$(1) \frac{1}{3}$$

$$(2) \frac{3}{4}$$

(3) $\frac{1}{2}$

(4) $\frac{2}{3}$

24. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \left(\frac{2r}{r^4 + r^2 + 2} \right) =$

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{2}$

(3) $-\frac{\pi}{4}$

(4) $-\frac{\pi}{2}$

25. $\operatorname{sech}^2 \left(\tanh^{-1} \frac{1}{2} \right) + \operatorname{cosech}^2 \left(\coth^{-1} 3 \right) =$

(1) $\frac{35}{9}$

(2) $\frac{3}{2}$

(3) $\frac{25}{4}$

(4) $\frac{35}{4}$

26. In ΔABC , if $a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$ then

(1) $2b = a + c$

(2) $b^2 = ac$

(3) $\frac{1}{b} = \frac{1}{a} + \frac{1}{c}$

(4) $a = c$

27. If s is the semi perimeter of ΔABC and if $\frac{s-a}{4} = \frac{s-b}{5} = \frac{s-c}{6}$

then $\sum \sin^2 \left(\frac{A}{2} \right) =$

(1) $\frac{74}{25}$

(2) $\frac{25}{74}$

(3) $\frac{74}{33}$

(4) $\frac{25}{33}$

28. If I is the incentre of $\triangle ABC$ and P_1, P_2, P_3 are respectively the radii of the circumcircles of the $\triangle IBC, \triangle ICA$ and $\triangle IAB$, then

$$P_1 P_2 P_3 =$$

(1) $2Rr$

(2) $2Rr^2$

(3) $2R^2r$

(4) $\frac{4R}{r}$

29. If $\vec{a}$ and $\vec{b}$ are unit vectors such that $\vec{a} + \vec{b}$ is also a unit vector,

then $|\vec{a} - \vec{b}|^2 =$

(1) 1

(2) 2

(3) 3

(4) 0

30. If the line joining the points $\hat{i} + \hat{j}$, and $3\hat{i} + \hat{j} - \hat{k}$ meets the plane that passes through the point $2\hat{i} + 4\hat{j}$ and parallel to the vector $3\hat{j} + 5\hat{k}$ and $3\hat{i} - \hat{k}$ at P , then the position vector of the point P is,

(1) $-27\hat{i} + \hat{j} + 14\hat{k}$

(2) $29\hat{i} + \hat{j} - 14\hat{k}$

(3) $-14\hat{i} + 89\hat{j} + 3\hat{k}$

(4) $2\hat{i} + 5\hat{j} - 7\hat{k}$

31. If A, B, C, D are four points in the plane such that

$$|\mathbf{AB}|^2 + |\mathbf{CD}|^2 = |\mathbf{BC}|^2 + |\mathbf{DA}|^2 = 100 \text{ then } \mathbf{AC} \cdot \mathbf{BD} =$$

(1) 10

(2) 0

(3) $\frac{1}{10}$

(4) 1

32. if $\mathbf{a} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}, \mathbf{b} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\mathbf{c}$ is a vector

perpendicular to $\mathbf{b}$ then,

$$\left| \left(\frac{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}{|\mathbf{b} \times \mathbf{c}|^2} \right) (\mathbf{b} \times \mathbf{c}) + \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \right) \mathbf{b} + \left(\frac{\mathbf{a} \cdot \mathbf{c}}{|\mathbf{c}|^2} \right) \mathbf{c} \right| =$$

(1) $\sqrt{14}$

(2) 14

(3) 13

(4) $\sqrt{17}$

33. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are the three vectors such that $|\mathbf{a}| = 1, |\mathbf{b}| = 2, |\mathbf{c}| = 3$ and

$$2\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{a} = 2 \text{ then } [\mathbf{a} \mathbf{b} \mathbf{c}]^2 =$$

(1) 15

(2) 14

(3) 12

(4) 8

34. If $\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}}, \mathbf{b} = \hat{\mathbf{j}} + \hat{\mathbf{k}}, \mathbf{c} = \hat{\mathbf{i}} + \hat{\mathbf{k}}$ and $\mathbf{c}$ is a vector perpendicular to

$$\mathbf{b} \text{ then, } \frac{[(\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a})] (\mathbf{c} \times \mathbf{a}) (\mathbf{a} \times \mathbf{b}) (\mathbf{a} \times \mathbf{b}) \times (\mathbf{b} \times \mathbf{c})}{[\mathbf{b} + \mathbf{c} \quad \mathbf{c} + \mathbf{a} \quad \mathbf{a} + \mathbf{b}][\mathbf{b} \times \mathbf{c} \quad \mathbf{c} \times \mathbf{a} \quad \mathbf{a} \times \mathbf{b}]}$$

- (1) 0
- (2) 1
- (3) $\sqrt{3}$
- (4) $\sqrt{2}$

35. The standard deviation of the scores 505, 510, 515, 520, ..., 595 is

- (1) $500 + 5\sqrt{30}$
- (2) $505 + \sqrt{30}$
- (3) $5\sqrt{30}$
- (4) $5 + \sqrt{30}$

36. If the variance of the distribution

x_i	4	8	11	17	20	24	32
f_i	3	5	9	5	4	3	1

is 45.8, then the variance of distribution

y_i	10	18	24	36	42	50	66
f_i	3	5	9	5	4	3	1

- (1) 93.6
- (2) $\sqrt{93.9}$

(3) 183.2

(4) $\sqrt{183.2}$

37. A die is formed so that the probability of getting a number i when it is rolled is proportional to i , ($i = 1, 2, 3, 4, 5, 6$). The probability of getting an odd number on the die when it is rolled is

(1) $\frac{1}{2}$

(2) $\frac{4}{7}$

(3) $\frac{2}{7}$

(4) $\frac{3}{7}$

38. A problem is given to 3 students A, B and C whose chances to solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Then, the probability of the problem being solved by exactly one of them, if all the three try independently, is

(1) $\frac{3}{4}$

(2) $\frac{11}{24}$

(3) $\frac{23}{24}$

(4) $\frac{1}{4}$

39. A certain recruitment test with multiple choice questions, there are four options to answer each question, out of which one is correct. An intelligent student knows 90% correct answers while a weak student known only 20% correct answers. If an intelligent student gets the correct answer for a question then the probability that he was guessing it, is

(1) $\frac{1}{37}$

(2) $\frac{1}{10}$

(3) $\frac{9}{37}$

(4) $\frac{1}{2}$

40. A random variable X has the following distribution

Values of	0	1	2	3	4	5	6	7
$X(x)$								
$P(X = x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Then $P(0 < X < 6) =$

(1) $\frac{9}{10}$

(2) $\left(\frac{9}{10}\right)^2$

(3) $\left(\frac{3}{10}\right)$

(4) $\frac{1}{10}$

41. An unbiased coin is tossed n times. If the probability of getting one head is greater than 0.8, then the least value of n is,

(1) 2

(2) 3

(3) 4

(4) 5

42. If a point P moves such that the sum of the distances from P to the point $A(1, -1)$ and $B(-1, 1)$ is always 4, then the equation for the locus of P is

(1) $16x^2 - 64x + 7y^2$

(2) $3x^2 + 2xy + 3y^2 = 8$

(3) $6x + 4y = 3$

(4) $x^2 + y^2 - 8x + 6y = 0$

43. The transformed equation of $3x^2 - 6xy + 8y^2 = 8$ when the axes are rotated about the origin through an angle $\frac{\pi}{4}$ in the positive direction is,

(1) $5x^2 + 10xy + 17y^2 + 16 = 0$

(2) $5x^2 + 10xy + 17y^2 - 16 = 0$

(3) $5x^2 - 10xy + 17y^2 - 16 = 0$

(4) $5x^2 - 10xy + 17y^2 + 16 = 0$

44. A variable line passing through a fixed point (α, β) intersects the coordinate axes at A and B. if O is the origin, then the locus of the centroid of the triangle OAB is

- (1) $\beta x + \alpha y - 2\alpha\beta = 0$
- (2) $\beta x + \alpha y - 3xy = 0$
- (3) $\alpha x + \beta y - (\alpha^2 + \beta^2) = 0$
- (4) $\beta x + \alpha y + 3xy = 0$

45. If $m = 1$ is the slope of a line L, then the product of the slopes of non parallel lines which are inclined at an angle of 60° with L is

- (1) 1
- (2) -1
- (3) $\sqrt{3}$
- (4) $-\frac{1}{2}$

46. The area (in sq units) of the quadrilateral formed by the lines

$$2x - 3y + 6 = 0, 2x + 3y + 6 = 0, 2x + 3y - 6 = 0 \text{ and}$$

$$2x - 3y - 6 = 0 \text{ is}$$

- (1) 12

(2) 36

(3) 6

(4) 18

47. If the straight lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$ forms a triangle with orthocentre at the origin, then $(a, b) =$

(1) $(-8, 8)$

(2) $(0, 7)$

(3) $(6, 4)$

(4) $(-3, 3)$

48. The product of the perpendicular distance from $(1, -1)$ to the pair of lines $x^2 - 4xy + y^2 = 0$ is

(1) 1

(2) $\frac{2}{3}$

(3) $\frac{3}{2}$

(4) 2

49. If the circle touches the line $3x - 4y - 10 = 0$ and $3x - 4y + 30 = 0$ and its centre lies on the line $x + 2y = 0$, then the equation of the circle is

(1) $x^2 + y^2 + 4x - 2y - 11 = 0$

(2) $x^2 + y^2 + 2x - 4y - 11 = 0$

(3) $x^2 + y^2 - 4x + 2y - 11 = 0$

(4) $x^2 + y^2 + 2x - y - 11 = 0$

50. If the line $4x + 4y - 11 = 0$ intersects the circle $x^2 + y^2 - 4x - 6y + 4 = 0$ at A and B , then the point of intersection of the tangents drawn at A , B is

(1) $(-1, 2)$

(2) $(-1, -2)$

(3) $(2, 1)$

(4) $(-2, -1)$

51. The equation of the circle which passes through the point $(3, 2)$ bisects the circumference of the circle $x^2 + y^2 = 15$ and cuts the circle $x^2 + y^2 + 4x + 6y + 3 = 0$ orthogonally is

(1) $x^2 + y^2 + 6x + 8y - 43 = 0$

(2) $x^2 + y^2 + 6x - 8y - 15 = 0$

(3) $x^2 + y^2 - 6x + 8y - 11 = 0$

(4) $x^2 + y^2 - 6x - 8y + 21 = 0$

52. $x^2 + y^2 + 2x + 4y - 20 = 0$ and $x^2 + y^2 + 6x - 8y + 10 = 0$ are given circles. Which one of the following is correct?

(1) They intersect orthogonally and will have two common tangents. The length of their common chord is $\frac{5\sqrt{3}}{\sqrt{2}}$

(2) They intersect at right angles and will have two common tangents. The length of their common chord is 2.

(3) They do not intersect orthogonally and will have three common tangents. The length of their direct chord is 5.

(4) They touch each other internally and will have only one common tangent.

53. The equation of the circle passing through the point of intersection of the circles $x^2 + y^2 + 4x + 6y - 12 = 0$ and $x^2 + y^2 - 6x - 4y - 12 = 0$ and cutting the circle $x^2 + y^2 - 4x + 4y + 8 = 0$ orthogonally is

(1) $x^2 + y^2 + 6x + 8y + 12 = 0$

(2) $x^2 + y^2 + 8x + 6y - 12 = 0$

(3) $x^2 + y^2 + 6x + 8y - 12 = 0$

(4) $x^2 + y^2 - 6x - 8y - 12 = 0$

54. If the parabola passes through the points $(-2, 1)$, $(1, 2)$ and $(-1, 3)$ having horizontal axis, then the length of the latus rectum of that parabola is

(1) 5

(2) $\frac{5}{2}$

(3) $\frac{2}{5}$

(4) $\frac{1}{5}$

55. The equation of one of the common tangents of the circle

$$x^2 + y^2 - 6y + 4 = 0 \text{ and the parabola } y^2 = x \text{ is}$$

(1) $2x - y + 1 = 0$

(2) $2x - y = 1$

(3) $4x - y + 1 = 0$

(4) $x - 2y + 1 = 0$

56. The equation of the ellipse having a vertex at $(6, 1)$, a focus at

$(4, 1)$ and the eccentricity $\frac{3}{5}$ is

(1) $\frac{(x-1)^2}{16} + \frac{(y-1)^2}{25} = 1$

(2) $\frac{(x-1)^2}{25} + \frac{(y-1)^2}{16} = 1$

(3) $\frac{(x+1)^2}{25} + \frac{(y+1)^2}{16} = 1$

(4) $\frac{(x+1)^2}{16} + \frac{(y+1)^2}{25} = 1$

57. If the tangent at the point $\left(4\cos\theta, \frac{16}{\sqrt{11}}\sin 2\theta\right)$ on the ellipse

$16x^2 + 11y^2 = 256$ and touches the circle $x^2 + y^2 - 2x = 15$, then

$\theta =$

(1) $\pm\frac{\pi}{3}$

(2) $\pm\frac{\pi}{6}$

(3) $\pm\frac{\pi}{4}$

(4) $\pm\frac{\pi}{8}$

58. A tangent to the curve $9b^2x^2 - 4a^2y^2 = 36a^2b^2$ makes intercept of unit length on each of the coordinate axis, then the point (a, b) lies on

(1) $x^2 - y^2 = 1$

(2) $x^2 + y^2 = 1$

(3) $4x^2 - 9y^2 = 1$

(4) $4x^2 + 9y^2 = 1$

59. The harmonic conjugate of $P(-9, 12, -15)$ with respect to the line segment AB, where $A = (1, -2, 3)$ and $B = (-4, 5, -6)$ is

(1) $\left(-\frac{2}{3}, \frac{1}{3}, 0\right)$

(2) $(6, -9, 12)$

(3) $\left(-\frac{7}{3}, \frac{8}{3}, -3\right)$

(4) $\left(\frac{7}{3}, -\frac{8}{3}, \frac{9}{3}\right)$

60. If the direction ratios of the lines L_1 and L_2 are $2, -1, 1$ and $3, -3, 4$ respectively, then the direction cosines of a line that is perpendicular to both L_1 and L_2 are

(1) $\pm \frac{2}{\sqrt{6}}, \pm \frac{1}{\sqrt{6}}, \pm \frac{1}{\sqrt{6}}$

(2) $\pm \frac{1}{\sqrt{35}}, \pm \frac{5}{\sqrt{35}}, \pm \frac{3}{\sqrt{35}}$

(3) $\pm \frac{3}{\sqrt{34}}, \pm \frac{3}{\sqrt{34}}, \pm \frac{4}{\sqrt{34}}$

(4) $\pm \frac{1}{\sqrt{14}}, \pm \frac{2}{\sqrt{14}}, \pm \frac{3}{\sqrt{14}}$

61. If the equation of the plane bisecting the line segment joining the point $P(3, 2, 4)$ and $Q(-1, 0, -2)$ and perpendicular to PQ is $ax + by + cz + d = 0$ then $ac + bd =$

- (1) 0
- (2) 12
- (3) 6
- (4) 1

62. $\cos \left[\lim_{x \rightarrow \infty} \frac{2\pi|x| + \pi x}{|x| - 3x} + \lim_{x \rightarrow 0} \frac{\cos\left(\frac{\pi}{2} \cos^2 x\right)}{x^2} \right] =$

- (1) 1
- (2) -1
- (3) 0
- (4) $\frac{1}{\sqrt{2}}$

63. $\lim_{n \rightarrow \infty} \frac{[6^2 + 12^2 + 18^2 + \dots + (6n)^2]^2}{[5 + 10 + 15 + \dots + 5n][2^3 + 4^3 + 6^3 + \dots + 8n^3]} =$

(1) $\frac{4}{5}$

(2) $\frac{144}{5}$

(3) $\frac{4}{25}$

(4) $\frac{144}{25}$

64. If $[x]$ denotes the greatest integer not exceeding the number x , is defined by

$$f(x) = \begin{cases} [x], & \text{if } x < 2 \\ [x] - 1 & \text{if } x \geq 2 \end{cases}$$

is continuous in the interval

(1) $(1, 2) \cup (2, 3)$

(2) $[1, 3)$

(3) $(1, 3)$

(4) R

65. If $a > 0$ and $f(x) = \left(\frac{a+x}{1+x}\right)^{a+1+2x}$, then $f'(0) =$

(1) a^{a+1}

(2) $a^{a+1} \left\{ \frac{1-a^2}{a} + 2\log a \right\}$

(3) $2\log a$

(4) $a^{a+1} \left\{ \frac{(1+a)^2}{a-2\log a} \right\}$

66. If $y = \log_2(\log_2 x)$, then $\frac{dy}{dx} =$

(1) $\frac{\log_2 e}{2x \log_e x}$

(2) $\frac{1}{x \log_e x \log_e 2}$

(3) $\frac{1}{\log_e (2x)^x}$

(4) $\frac{1}{\log_2 e \log_e x}$

67. If $y = x \log \left(\frac{x}{2-3x} \right)$, for $0 < x < \frac{2}{3}$ then $\frac{d^2 y}{dx^2}$ at $x = \frac{1}{2}$ is

(1) 4

(2) 16

(3) 32

(4) 2

68. An approximate value of $\sqrt[4]{18}$ is

(1) 2.0512

(2) 2.0425

(3) 2.0625

(4) 2.0834

69. The sum of the maximum and minimum values of

$f(x) = 3x^4 - 2x^3 - 6x^2 + 6x + 4$ in $(0, 2)$ is

(1) 28

(2) $\frac{167}{16}$

(3) $\frac{134}{15}$

(4) $\frac{87}{16}$

70. The constant c of Lagrange's mean value theorem for

$$f(x) = \cos x - \sin 2x \text{ in } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \text{ is}$$

(1) 0

(2) $\sin^{-1}\left(\frac{1 \pm \sqrt{33}}{8}\right)$

(3) $\cos^{-1}\left(\frac{1 + \sqrt{33}}{8}\right)$

(4) $\pm \frac{\pi}{4}$

71. Air is discharging from a large spherical balloon at the rate of 4 cubic meters per minute. Then the rate at which the surface area is shrinking when the radius of balloon is 8 m is,

(1) $2 \text{ m}^2/\text{minute}$

(2) $1 \text{ m}^2/\text{minute}$

(3) $4 \text{ m}^2/\text{minute}$

(4) $8 \text{ m}^2/\text{minute}$

$$72. \int \frac{x}{(x-1)^2(x+2)} dx =$$

$$(1) \frac{2}{9} \log \left| \frac{x-1}{x+2} \right| + c$$

$$(2) \frac{2}{9} \log \left| \frac{x+2}{x-1} \right| + c$$

$$(3) \frac{1}{3} \left(\frac{1}{(x-1)} \right) + \frac{2}{9} \log \left| \frac{x-1}{x+2} \right| + c$$

$$(4) -\frac{1}{3} \left(\frac{1}{(x-1)} \right) + \frac{2}{9} \log \left| \frac{x-1}{x+2} \right| + c$$

$$73. \int \frac{\sqrt{x^2+1} [\log(x^2+1) - 2\log x]}{x^4} dx =$$

$$(1) \frac{1}{9} \left(1 + \frac{1}{x^2} \right)^{\frac{3}{2}} \left[2 - 3 \log \left(1 + \frac{1}{x^2} \right) \right] + c$$

$$(2) \frac{1}{3} \left(1 + \frac{1}{x^2} \right)^{\frac{1}{2}} \left[6 - \log \left(1 + \frac{1}{x^2} \right)^2 \right] + c$$

$$(3) \frac{1}{9} \left(1 + \frac{1}{x^2} \right) \left[3 - 2 \log \left(1 + \frac{1}{x^2} \right)^{\frac{1}{2}} \right] + c$$

$$(4) \frac{1}{3} \left(1 + \frac{1}{x^2} \right)^{\frac{3}{2}} \left[3 + \log \left(1 + \frac{1}{x^2} \right) \right] + c$$

$$74. \int \frac{dx}{\sin x + \sin 2x} =$$

$$(1) \frac{1}{6} \log(1 - \cos x) + \frac{1}{2} \log(1 + \cos x) + \frac{2}{3} \log|1 + 2 \cos x| + c$$

$$(2) \frac{1}{6} \log(1 - \cos x) - \frac{1}{2} \log(1 + \cos x) - \frac{2}{3} \log|1 + 2 \cos x| + c$$

$$(3) \frac{1}{6} \log(1 - \cos x) + \frac{1}{2} \log(1 + \cos x) - \frac{2}{3} \log|1 + 2 \cos x| + c$$

$$(4) \frac{1}{6} \log[(1 - \cos x)(1 + \cos x)|1 + 2 \cos x|] + c$$

$$75. \int (\sec^4 x + \tan^4 x) dx =$$

$$(1) \frac{2}{3} \tan^3 x - \frac{2}{3} \tan x + x + c$$

$$(2) \frac{1}{3} \sec^2 x \tan x + \frac{5}{3} \tan x + \frac{\tan^3 x}{3} + x + c$$

$$(3) \frac{2}{3} \tan^3 x + x + c$$

$$(4) \frac{1}{3} \sec^2 x \tan x - \frac{5}{3} \tan x + \frac{\tan^3 x}{3} + x + c$$

$$76. \lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \sin^5 \left(\frac{\pi}{6n} \right) + \sin^5 \left(\frac{2\pi}{6n} \right) + \sin^5 \left(\frac{3\pi}{6n} \right) + \dots + \sin^5 \left(\frac{\pi}{2} \right) \right\} =$$

$$(1) \frac{8}{15\pi}$$

$$(2) \frac{8}{5\pi}$$

$$(3) \frac{32}{5\pi}$$

$$(4) \frac{16}{5\pi}$$

$$77. \int_0^{\frac{\pi}{2}} \log_e (\sin 2x) dx =$$

$$(1) \pi \log 2$$

$$(2) -\pi \log 2$$

$$(3) \frac{\pi}{2} \log 2$$

$$(4) -\frac{\pi}{2} \log 2$$

78. The area in (sq units) between the curves $y^2 = 8x$ and its latus rectum is

- (1) $\frac{32}{3}$
- (2) $\frac{64}{3}$
- (3) $\frac{16}{3}$
- (4) $\frac{8\sqrt{2}}{3}$

79. The order and degree of the differential equation

$y = px + \sqrt{a^2 p^2 + b^2}$ are respectively

- (1) 2, 1
- (2) 1, 2
- (3) 1, 1
- (4) 2, 2

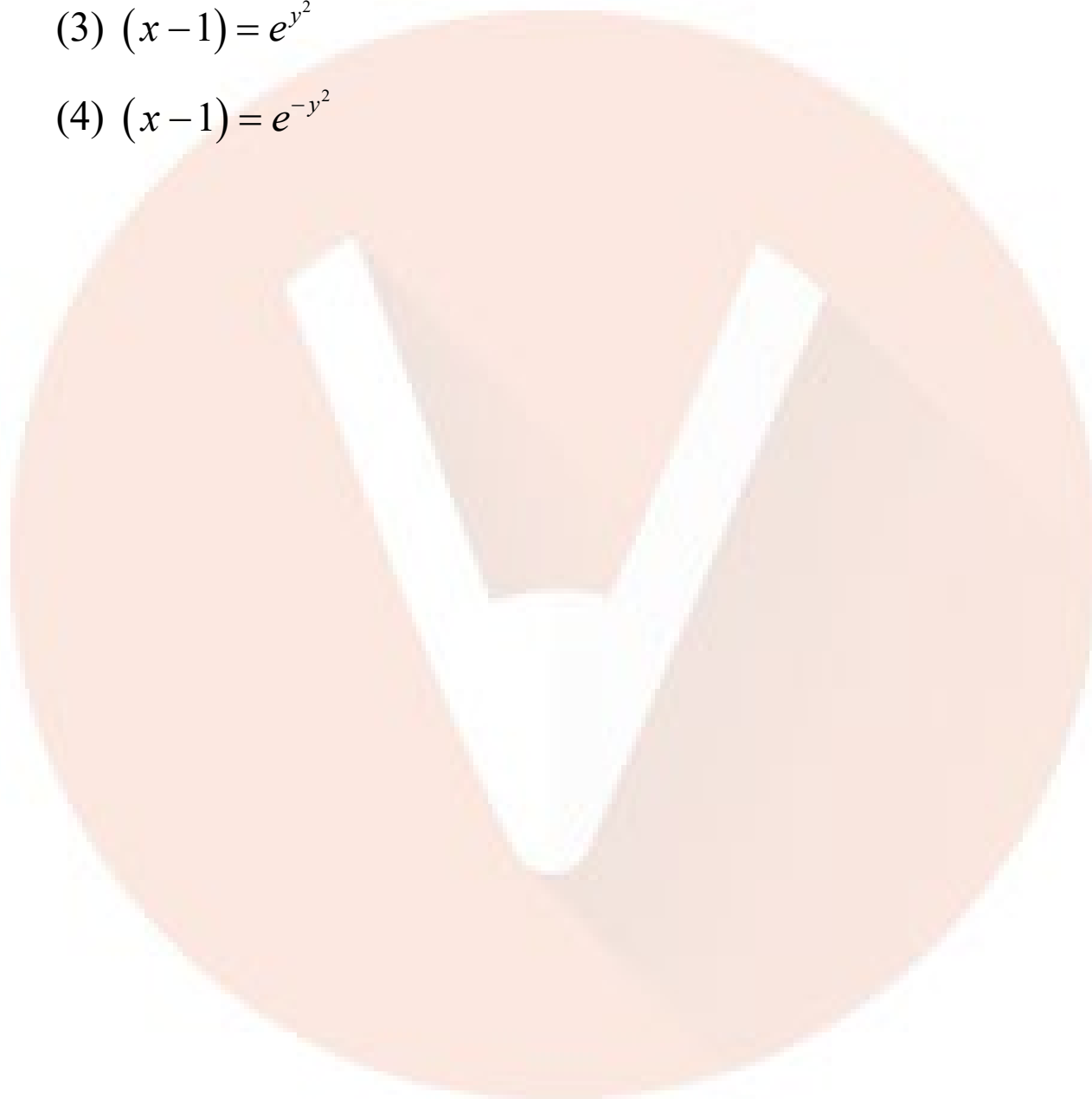
80. The solution of the differential equation $\frac{dx}{dy} + 2yx = 2y$ which passes through the point $(2, 0)$ is

$$(1) (x-1) = 2e^{y^2}$$

$$(2) (x-1) = 2e^{-y^2}$$

$$(3) (x-1) = e^{y^2}$$

$$(4) (x-1) = e^{-y^2}$$



AP EAMCET-2018-Engineering-23 Apr-Shift-2 Physics

81. Match the measurements given in List-I with the number of significant figures given in List-II.

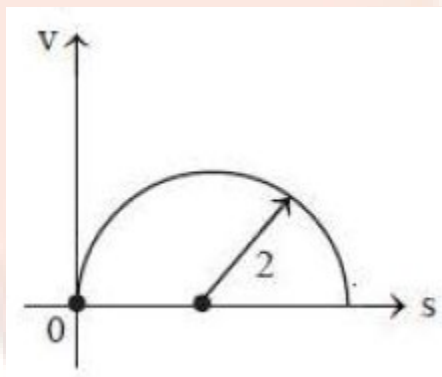
<u>List-I</u>		<u>List-II</u>	
(A)	74.083	(I)	3
(B)	0.029	(II)	4
(C)	0.002407	(III)	2
(D)	2.74×10^7	(IV)	5

The correct answer is

	A	B	C	D
(1)	IV	II	III	I
	A	B	C	D
(2)	IV	III	II	I
	A	B	C	D
(3)	III	IV	II	I
	A	B	C	D
(4)	I	II	III	IV

82. The velocity-displacement (v-s) graph shows the motion of particle moving in a straight line. Velocity-displacement graph

is a circle of radius 2 m and centre is at (2, 0) m. The value of acceleration for this particle at a point $(2 - \sqrt{2}, \sqrt{2})$ m will be _____ ms^{-2} .



- (1) $\sqrt{2}$
- (2) 4
- (3) 2
- (4) 3

83. A body is projected horizontally from the top of a tall tower with a velocity of 30 ms^{-1} . At time t_1 , its horizontal and vertical components of the velocity are equal and at t_2 , its horizontal and vertical displacements are equal. Then $(t_2 - t_1) =$

$(g = 10 \text{ ms}^{-2})$

- (1) 1 s
- (2) 1.5 s

(3) 2 s

(4) 3 s

84. A particle is projected at an angle of 60° with the horizontal from the ground with a velocity $10\sqrt{3} \text{ ms}^{-1}$. The angle between velocity vector after 2 s and initial velocity vector is ($g=10 \text{ ms}^{-2}$)

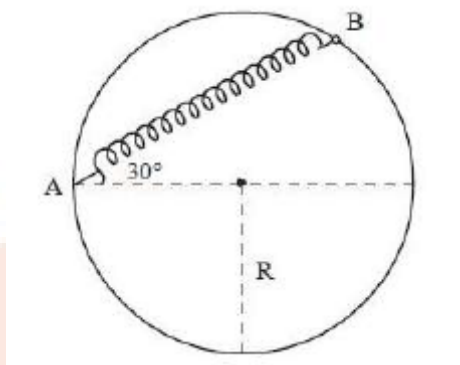
(1) 0°

(2) 30°

(3) 60°

(4) 90°

85. A bead of mass 100 g is attached to one end of a spring of natural length L and spring constant $k = \frac{(\sqrt{3} + 1)mg}{L}$, where m is the mass of bead. The other end of the spring is fixed at point A on a smooth vertical ring of radius R as shown in the figure. The normal reaction at B just after it is released to move is ($g = 9.8 \text{ ms}^{-2}$)



- (1) 1.73 N
- (2) 2.23 N
- (3) 2.44 N
- (4) 2.55 N

86. A rocket with an initial mass ' m_0 ' is going up with a constant acceleration ' a ' by exhausting gases with a velocity ' v ' relative to the rocket motion, then the mass of the rocket at any instant of time is (Assume that no other forces act on it)

- (1) $m = m_0 e^{-\frac{a}{v}t}$
- (2) $m = m_0 e^{-\frac{2a}{v}t}$
- (3) $m = m_0 e^{-\frac{a}{2v}t}$
- (4) $m = m_0 e^{-\frac{a^2}{v^2}t^2}$

87. A particle is released freely from a height H . At a certain height, its kinetic energy is two times its potential energy. Then the height and the speed of the particle at that instant are respectively (g = acceleration due to gravity)

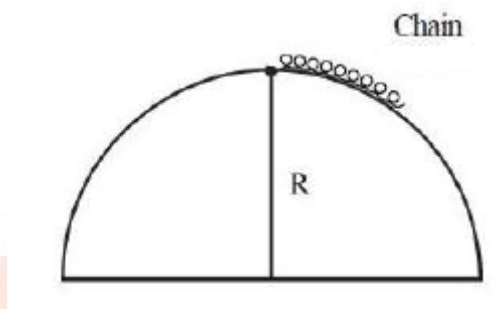
(1) $\frac{H}{3}, \sqrt{\frac{2gH}{3}}$

(2) $\frac{H}{3}, 2\sqrt{\frac{gH}{3}}$

(3) $\frac{2H}{3}, \sqrt{\frac{2gH}{3}}$

(4) $\frac{H}{3}, \sqrt{2gH}$

88. A uniform chain of length ' l ' and mass ' m ' lies on the surface of a smooth hemisphere of radius R ($R > l$) with one end tied to the top of the hemisphere as shown in the figure. Gravitational potential energy of the chain with respect to the base of the hemisphere is



(1) $\frac{mgl}{2}$

(2) $\frac{mgR^2}{l} \sin\left(\frac{l}{R}\right)$

(3) $\frac{mgR^2}{l} \sin\left(\frac{R}{l}\right)$

(4) $\frac{mgl^2}{R} \sin\left(\frac{l}{R}\right)$

89. A particle of mass 15 kg is moving with a uniform speed of 8 ms^{-1} in $x - y$ plane along the line $3y = 4x + 10$, then the magnitude of its angular momentum about the origin in $\text{kg} \cdot \text{m}^2 \text{s}^{-1}$ is _____ $\left(\sin 53^\circ = \frac{4}{5}\right)$

(1) 240

(2) 80

(3) 120

(4) 280

90. An empty bucket of mass 1 kg attached by a light cord passed over a pulley of a water well is released from rest. If the pulley assembly is assumed to be a uniform solid cylinder of mass 8 kg and free to rotate about its axis without any friction, then the speed of the bucket as it hits the water 16 m below is

($g = 10 \text{ ms}^{-2}$)

(1) 4 ms^{-1}

(2) 8 ms^{-1}

(3) 16 ms^{-1}

(4) 20 ms^{-1}

91. The displacement of a particle of mass 2 g executing SHM is given by $y = 5 \sin\left(4t + \frac{\pi}{3}\right)$. Here y is in meters and t is in seconds. The kinetic energy of the particle when $t = \frac{T}{4}$ is

(1) 0.4 J

(2) 0.5 J

(3) 3 J

(4) 0.3 J

92. Two bodies of equal masses are some distance apart. If 20% of mass is transferred from the first body to the second body, the gravitational force between them

- (1) Increases by 4%
- (2) Increases by 14%
- (3) Decreases by 4%
- (4) Decreases by 14%

93. One end of a long metallic wire of length L , area of cross-section A and Young's modulus Y is tied to the ceiling. The other end is tied to a massless spring of force constant K and a mass ' m ' is hung from the free end of the spring. If ' m ' is slightly pulled down and released, then its time period of oscillation is

- (1) $2\pi\sqrt{\frac{m}{K}}$
- (2) $2\pi\sqrt{\frac{mYA}{KL}}$

$$(3) \ 2\pi\sqrt{\frac{m(KA + YL)}{KYA}}$$

$$(4) \ 2\pi\sqrt{\frac{m(KL + YA)}{KYA}}$$

94. Two solid spheres of radii 2 mm and 4 mm are tied to the two ends of a light string and released in a liquid of specific gravity 1.3 and coefficient of viscosity 1 Pa s. The string is just taut when the two spheres are completely in the liquid. If the density of the materials of the two spheres is 2800 kgm^{-3} , the terminal velocity of the system of the spheres is

$$(g = 10 \text{ ms}^{-2})$$

$$(1) \ 2 \text{ cms}^{-1}$$

$$(2) \ 4 \text{ cms}^{-1}$$

$$(3) \ 4 \text{ ms}^{-1}$$

$$(4) \ 2 \text{ ms}^{-1}$$

95. Assertion (A) : A room can be cooled by opening the door of a refrigerator in it.

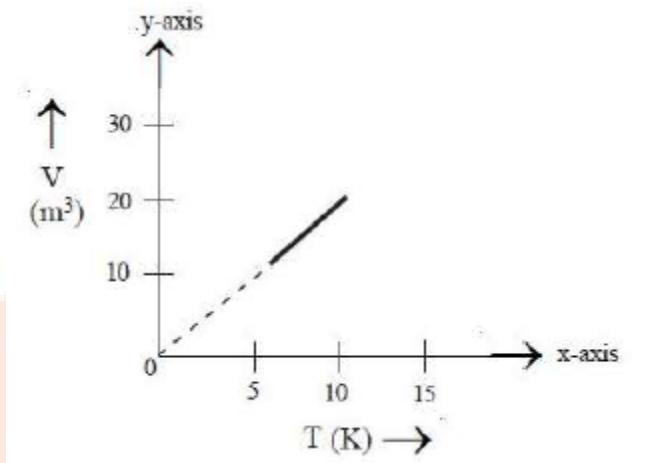
Reason (R) : Heat always flows from a body at higher temperature to a body at lower temperature.

- (1) (A) and (R) are true; (R) is the correct explanation of (A)
- (2) (A), (R) are true; (R) is not the correct explanation of (A)
- (3) (A) is true, (R) is false.
- (4) (A) is false, (R) is true.

96. A wire of $20\ \Omega$ is immersed in ice. If 10 A current is passed through this wire for 1 minute, ice completely melts. The mass of the ice is nearly ($L_{ice} = 79.7\ \text{calg}^{-1}$)

- (1) 3.5 g
- (2) 359 g
- (3) 540 g
- (4) 3.5 kg

97. A graph drawn between absolute temperature and volume of 3 moles of helium gas as shown in the figure. If 5 cal of heat is used in the process, then the work done is



- (1) 21.0 J
- (2) 8.4 J
- (3) 12.6 J
- (4) 6.2 J

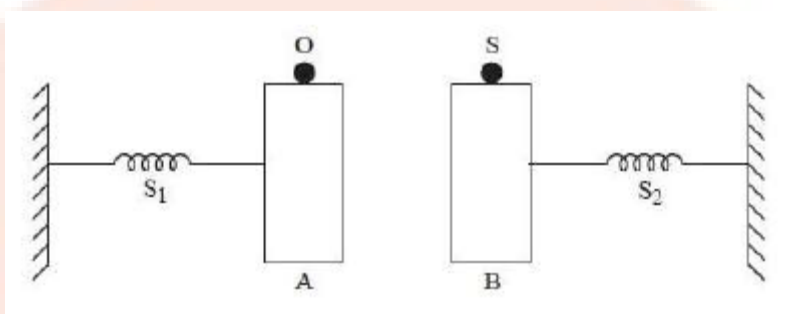
98. An ideal gas is found to obey $PV^{\frac{3}{2}} = \text{constant}$ during an adiabatic process. If such a gas initially at a temperature T is adiabatically compressed to half of its initial volume, then its final temperature is

- (1) $\sqrt{2}T$
- (2) $2T$
- (3) $2\sqrt{2}T$
- (4) $4T$

99. The rms speed of oxygen molecule at a certain temperature is 600 ms^{-1} . If the temperature is doubled and oxygen molecules dissociates into atomic oxygen atoms, the new rms speed is
- (1) 120 ms^{-1}
 - (2) 150 ms^{-1}
 - (3) 1200 ms^{-1}
 - (4) 600 ms^{-1}
100. A progressive wave of frequency 500 Hz is travelling with a velocity of 360 ms^{-1} . The distance between the two points, having a phase difference of 60° , is _____
- (1) 1.2 m
 - (2) 12 m
 - (3) 0.12 m
 - (4) 0.012 m
101. A source S emitting sound of frequency 288 Hz is fixed on block B which is attached to the free end of a spring S_2 and an observer O is on block A which is attached to the free end of spring S_1 ; as shown in figure. The blocks A and B are simultaneously displaced towards each other through a distance

of 0.5 m and then left to oscillate. If the angular velocity of each block is 40 rads^{-1} , the maximum frequency observed by the observer is

(speed of sound in air is 340 m s^{-1})



- (1) 288 Hz
- (2) 310 Hz
- (3) 324 Hz
- (4) 256 Hz

102. In a compound microscope, the focal lengths of two lenses are 1.5 cm and 6.25 cm. An object is placed at 2 cm from the objective and final image is formed at 25 cm from the eye lens. The distance between the two lenses is _____ (in cm)

- (1) 6
- (2) 7.75
- (3) 9.25
- (4) 11

103. In Young's double slit experiment, the intensity of central fringe is I_0 and fringe width is β . If a point is at a distance x from the central fringe, the intensity at that point is

(1) $I_0 \cos^2 \left(\frac{\pi x}{\beta} \right)$

(2) $I_0 \cos^2 \left(\frac{x}{\beta} \right)$

(3) $\frac{I_0}{4} \cos^2 \left(\frac{\pi x}{\beta} \right)$

(4) $I_0 \cos^2 \left(\frac{\pi \beta}{x} \right)$

104. A proton and an α -particle start from rest in a uniform electric field. The ratio of times taken by them to travel the same distance in the field is _____

(1) $\sqrt{5} : \sqrt{2}$

(2) $\sqrt{3} : 1$

(3) $2 : 1$

(4) $1 : \sqrt{2}$

105. Two charged balls moving in the same direction with same velocity v are placed in an electric field. After some time, one ball moves with velocity $\frac{v}{2}$ at an angle of 60° with the initial direction and the other ball moves at right angles to the initial direction with a velocity v' . Then the value of v' is

(1) $\frac{v}{\sqrt{2}}$

(2) $\frac{v}{\sqrt{3}}$

(3) $\frac{v}{2}$

(4) v

106. Electric field vector in a region is given by $\vec{E} = (3\hat{i} + 4y\hat{j}) \text{ Vm}^{-1}$

. The potential at the origin is zero. Then the potential at a point

(2, 1) m is

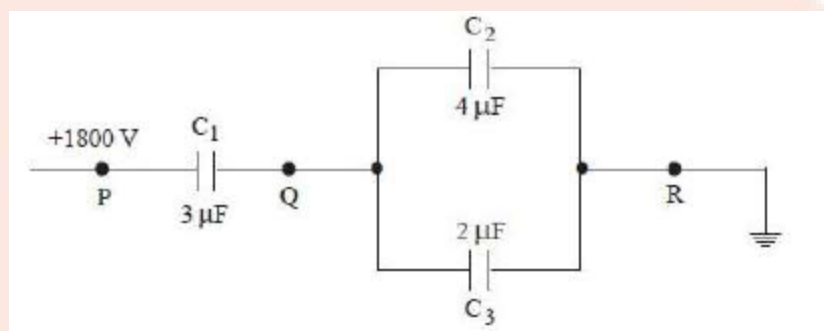
(1) 7 V

(2) 8 V

(3) -8 V

(4) -7 V

107. In the circuit shown in figure, if the point R is earthed and point P is given a potential of $+1800\text{ V}$, then charges on C_2 and C_3 are respectively _____



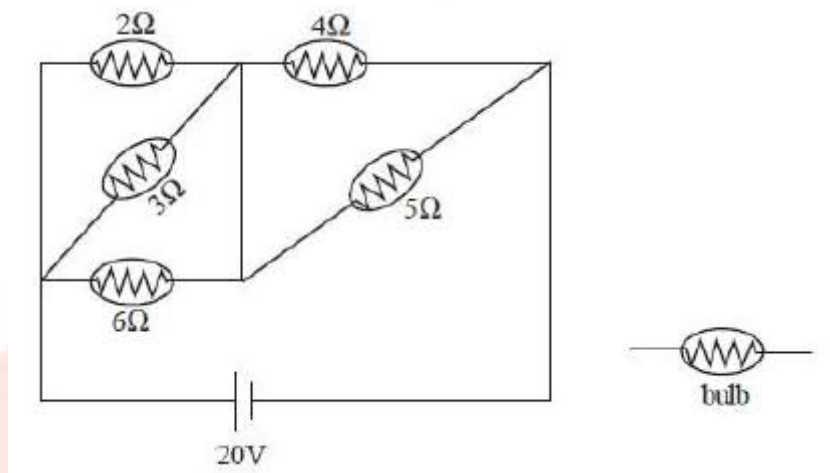
(1) $2.4 \times 10^{-3}\text{ C}$; $1.2 \times 10^{-3}\text{ C}$

(2) $1.6 \times 10^{-3}\text{ C}$; $0.8 \times 10^{-3}\text{ C}$

(3) $3.2 \times 10^{-3}\text{ C}$; $1.6 \times 10^{-3}\text{ C}$

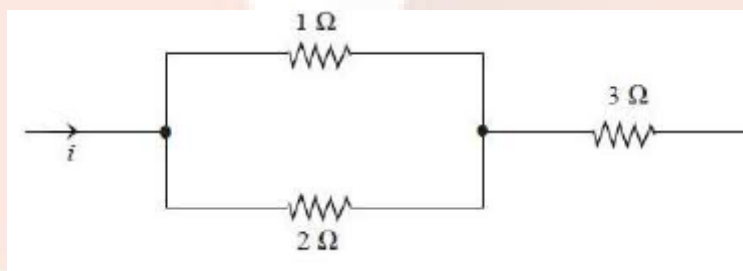
(4) $4.8 \times 10^{-3}\text{ C}$; $2.4 \times 10^{-3}\text{ C}$

108. The bulb which glows with maximum intensity in the given circuit is



- (1) 4 Ω bulb
- (2) 2 Ω bulb
- (3) 3 Ω bulb
- (4) 6 Ω bulb

109. In the circuit shown in figure, power developed across 1 Ω , 2 Ω and 3 Ω resistances are in the ratio



- (1) 1 : 2 : 3
- (2) 4 : 2 : 27
- (3) 6 : 4 : 9

(4) 2 : 1 : 27

110. Two long straight parallel conductors are carrying currents i_1 and i_2 in the same direction. Work done per unit length when the distance between them is doubled is

(1) $2 \times \frac{\mu_0}{2\pi} i_1 i_2$

(2) $\frac{\mu_0}{2\pi} i_1 i_2 \ln[2]$

(3) $\frac{\mu_0}{2\pi} i_1 i_2 \ln[4]$

(4) 0

111. A straight conductor of length 32 cm carries a current of 30 A. Magnetic induction at a point in air at a perpendicular distance of 12 cm from the midpoint of the conductor is _____

(1) 0.2 gauss

(2) 0.3 gauss

(3) 0.4 gauss

(4) 0.5 gauss

112. A sample of a paramagnetic salt containing 3×10^{24} atomic dipoles each of dipole moment $2 \times 10^{-23} \text{ Am}^2$ is subjected to a uniform magnetic field of 880 mT and cooled to a temperature of 3.5 K. The degree of magnetic saturation achieved is 10%. If the sample is subjected to a magnetic field of 990 mT and cooled to a temperature of 2.1 K, the total dipole moment of the sample is

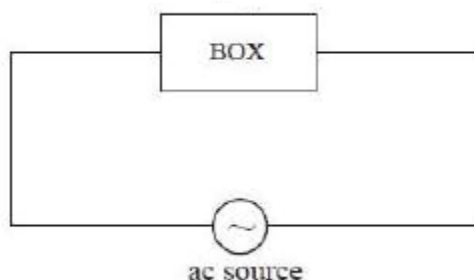
- (1) 11.25 Am^2
- (2) 23.5 Am^2
- (3) 15 Am^2
- (4) 75 Am^2

113. A coil of wire of radius ' r ' has 600 turns and self inductance of 108 mH. The self inductance of a coil with same radius and 500 turns is _____

- (1) 80 mH
- (2) 75 mH
- (3) 108 mH
- (4) 90 mH

114. In the AC circuit shown, $E = E_0 \sin(\omega t + \phi)$ and

$i = i_0 \sin\left(\omega t + \phi + \frac{\pi}{4}\right)$. Then the box contains



- (1) Only C
- (2) L and R in series
- (3) C and R in series or L , C and R in series
- (4) Only R

115. The oscillating electric field of an electromagnetic wave is given by $E_y = 30 \sin(2 \times 10^{11} t + 300 \pi x) \text{ Vm}^{-1}$. Then the value of wavelength of the electromagnetic wave is _____

- (1) $5.67 \times 10^{-3} \text{ m}$
- (2) $6.67 \times 10^{-3} \text{ m}$
- (3) $66.7 \times 10^{-3} \text{ m}$
- (4) $7.66 \times 10^{-3} \text{ m}$

116. Photons of wavelength λ emitted by a source of power P incident on a photo cell. If the current produced in the cell is I . The percentage of incident photons which produce current in the photo cell is

(h is Planck's constant and c is the speed of light in vacuum)

(1) $\frac{100ePc}{Ih\lambda}$

(2) $\frac{100eP\lambda}{Ihc}$

(3) $\frac{100Ih\lambda}{ePc}$

(4) $\frac{100Ihc}{eP\lambda}$

117. If λ_1 and λ_2 are the wavelength of the photons emitted when electrons in the n^{th} orbit of hydrogen atom fall to first excited state and ground state respectively, then the value of n is

(1) $\sqrt{\frac{2(\lambda_2 - \lambda_1)}{2\lambda_2 - \lambda_1}}$

$$(2) \sqrt{\frac{2\lambda_2 - \lambda_1}{2(\lambda_2 - \lambda_1)}}$$

$$(3) \sqrt{\frac{4\lambda_2 - \lambda_1}{4(\lambda_2 - \lambda_1)}}$$

$$(4) \sqrt{\frac{4(\lambda_2 - \lambda_1)}{(4\lambda_2 - \lambda_1)}}$$

118. The number of half lives elapsed before 93.75% of a radioactive sample has decayed is

- (1) 6
- (2) 4
- (3) 2
- (4) 8

119. In the common base configuration, a transistor has current amplification factor 0.95. If the transistor is used in common emitter configuration and the base current changes by $2\mu\text{A}$, then the change in the collector current is

- (1) $19\mu\text{A}$
- (2) $0.91\mu\text{A}$

(3) $1.9 \mu\text{A}$

(4) $38 \mu\text{A}$

120. If the height of the transmitting tower is increased by 30%, the area covered by it increases by

(1) 10%

(2) 21%

(3) 30%

(4) 60%

AP EAMCET-2018-Engineering-23 Apr-Shift-2 Chemistry

121. When a metal surface is exposed to certain frequency of electromagnetic radiation, the kinetic energy of electron ejected from metal surface is 0.20 eV. If its work function (W_0) is 4.80 eV, the approximate frequency of radiation falling on the metal surface in Hz is

- (1) 1.98×10^{15}
- (2) 1.21×10^{16}
- (3) 1.21×10^{15}
- (4) 1.98×10^{16}

122. If the ratio of energies of electron in the excited states of H and Li^{2+} is 1:9, the radius ratio of electron in the same excited states of H and Li^{2+} is

- (1) 9:1
- (2) 3:1
- (3) 1:9
- (4) 1:3

123. Identify the correct statements from the following.

- i. In the periodic table, about 78% of elements are metals.
- ii. In a group, the metallic character decreases from top to bottom and in a period the non-metallic character decreases from left to right.
- iii. The element Ho belongs to f-block.

- (1) i, ii, iii
- (2) ii, iii
- (3) i, iii
- (4) i, ii

124. The correct order of dipole moments of NH_3 , H_2O and NF_3 is

- (1) $\text{H}_2\text{O} > \text{NH}_3 > \text{NF}_3$
- (2) $\text{H}_2\text{O} > \text{NF}_3 > \text{NH}_3$
- (3) $\text{NF}_3 > \text{NH}_3 > \text{H}_2\text{O}$
- (4) $\text{NH}_3 > \text{NF}_3 > \text{H}_2\text{O}$

125. The number of electrons present in bonding and antibonding orbitals in O_2^{2-} is respectively

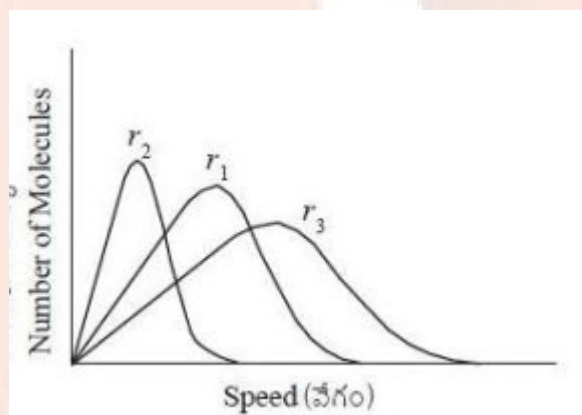
- (1) 10, 6

(2) 12, 6

(3) 11, 7

(4) 10, 8

126. If r_1 , r_2 and r_3 represent the most probable speeds of three different gases at the same temperature as shown in figure with molar masses M_1 , M_2 and M_3 respectively, the correct order of molar masses of these gases is



(1) $M_1 > M_3 > M_2$

(2) $M_3 > M_2 > M_1$

(3) $M_2 > M_1 > M_3$

(4) $M_2 > M_3 > M_1$

127. The volume of 0.1 M HCl required in mL to neutralise 20 mL of a solution containing 0.106 g of Na_2CO_3 is

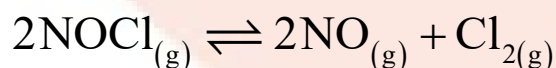
- (1) 10
- (2) 5
- (3) 20
- (4) 40

128. If enthalpy of combustion of carbon to $\text{CO}_{2(g)}$ is

$-394.0 \text{ kJ mol}^{-1}$, the enthalpy change for the formation of 17.6 g of CO_2 from carbon and dioxygen at the same temperature in kJ is

- (1) -157.6
- (2) 315.2
- (3) 157.6
- (4) -315.2

129. At 1000 K, if the equilibrium constant K_p for the reaction



is 4.157×10^{-4} bar, the K_c (in mol L^{-1}) is

$$(R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1})$$

- (1) 4.16×10^{-7}

(2) 4.16×10^{-4}

(3) 5.0×10^{-4}

(4) 5.0×10^{-6}

130. If the ionization constant of hypochlorous acid (HOCl) is 2.5×10^{-5} , the pH of 1.0 M of its solution is ($\log 5 = 0.7$)

(1) 3.3

(2) 2.3

(3) 4.3

(4) 3.0

131. In which of the following reactions, oxygen is not liberated?

(1) Reaction of HOCl with H_2O_2

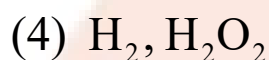
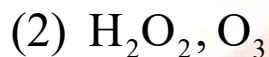
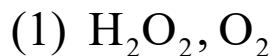
(2) Reaction of acidified KMnO_4 with H_2O_2

(3) Reaction of iodine with H_2O_2 in basic medium

(4) Reaction of lead sulphide with H_2O_2

132. A compound (M_2O_2) of group I element (M) hydrolyses to form M^+ , OH^- and X. Another compound ($\text{M}'\text{O}_2$) of group I

element (M') hydrolyses to form $(M')^+$, OH^- , X and Y. What are X and Y respectively?

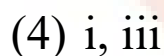
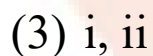
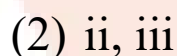
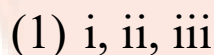


133. Identify the correct statements from the following

i. The atomic radius of Al is lower than the atomic radius of Ga.

ii. Boron exists in many allotropic forms.

iii. The melting point of Ga is lowest among the group 13 elements.

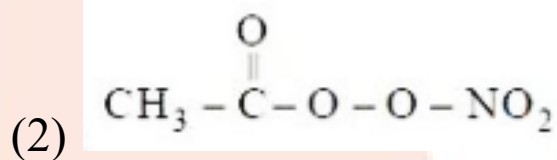


134. Which of the following is not correct corresponding to chemistry of group 14 elements?

- (1) Lead has no reaction with water due to formation of protective oxide layer.
- (2) GeX_2 is more stable than GeX_4
- (3) PbX_2 is more stable than PbX_4
- (4) Tin on reaction with steam liberates hydrogen

135. The chemical substance of photochemical smog responsible for eye irritation is

- (1) $\text{CH}_2 = \text{CH} - \text{CHO}$



- (3) $\text{CH}_2 = \text{CH}_2$
- (4) CH_4

136. Match the following

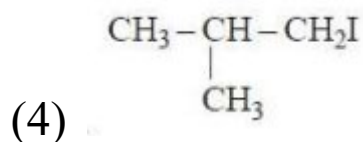
List-I	List-II
A) Resonance	I) $>\text{C}=\text{C}< + \text{H}^{\oplus} \longrightarrow \begin{array}{c} >\text{C}-\overset{\oplus}{\text{C}}< \\ \\ \text{H} \end{array}$
B) Inductive effect	II)

	$\text{H}-\text{CH}_2-\overset{\oplus}{\text{C}}\text{H}_2 \leftrightarrow \overset{\oplus}{\text{H}}\text{CH}_2=\text{CH}_2$
C) Electromeric effect	III) C_6H_6
D) Hyperconjugation	IV) $\text{CH}_3-\text{Z} \longrightarrow \overset{\ominus}{\text{C}}\text{H}_3 + \overset{\oplus}{\text{Z}}$
	V) $\text{CH}_3-\text{CH}_2-\text{CH}_2\text{Cl}$

- | | | | | |
|-----|-----|-----|----|-----|
| | A | B | C | D |
| (1) | II | I | IV | III |
| | A | B | C | D |
| (2) | III | V | I | II |
| | A | B | C | D |
| (3) | I | III | II | V |
| | A | B | C | D |
| (4) | III | II | I | IV |

137. The rate of dehydrohalogenation of which one among the following is less?

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
- (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
- (3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{I}$



138. Propyne reacts with HBr to form Z. The compound Z is

- (1) $\text{CH}_2(\text{Br})\text{CH}_2\text{CH}_2\text{Br}$
- (2) $\text{CH}_3\text{CH}_2\text{CHBr}_2$
- (3) $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{Br}$
- (4) $\text{CH}_3\text{CBr}_2\text{CH}_3$

139. A metal oxide crystallises in a hexagonal close-packed array of oxide ions with two out of every three octahedral holes occupied by metal ions. The formula of metal oxide is

- (1) MO
- (2) M_3O_4
- (3) M_2O_5
- (4) M_2O_3

140. The quantity of CO_2 in 500 mL of soda water when packed under 3.34 bar CO_2 pressure at 298 K in g is

(Henry's law constant for CO_2 in water at 298 K is 1.67×10^8 Pa)

- (1) 2.442
- (2) 1.221
- (3) 4.884
- (4) 3.663

141. 300 mL of an aqueous solution of a protein contains 2.52 g of the protein. If osmotic pressure of such a solution at 300 K is 5.04×10^{-3} bar, the molar mass of the protein in g mol^{-1} is

- (1) 83.0×10^3
- (2) 20.8×10^3
- (3) 41.5×10^3
- (4) 41.5×10^4

142. The conductivity of 0.01 M aqueous acetic acid measured with a conductivity cell of cell constant of 0.5 cm^{-1} at 298 K is $3.12 \times 10^{-4} \text{ S}$. If the limiting conductivities of H^+ and CH_3COO^- at the same temperature are 349, and $41 \text{ S cm}^2 \text{ mol}^{-1}$ respectively, the dissociation constant of acetic acid is

- (1) 1.67×10^{-4}

(2) 1.67×10^{-5}

(3) 1.67×10^{-3}

(4) 1.67×10^{-6}

143. At T(K), the following data were obtained for a general reaction $A + B + C \rightarrow \text{products}$

Expt.	Initial[A]	Initial[B]	Initial[C]	Initial rate
1	0.02 M	0.1 M	0.03 M	$2.4 \times 10^{-6} \text{ Ms}^{-1}$
2	0.02 M	0.2 M	0.03 M	$4.8 \times 10^{-6} \text{ Ms}^{-1}$
3	0.02 M	0.2 M	0.06 M	$9.6 \times 10^{-6} \text{ Ms}^{-1}$
4	0.02 M	0.2 M	0.06 M	$9.6 \times 10^{-6} \text{ Ms}^{-1}$

The rate constant for the above reaction is

(1) $8.0 \times 10^{-4} \text{ s}^{-1}$

(2) $8.0 \times 10^{-4} \text{ L mol}^{-1} \text{ s}^{-1}$

(3) $8.0 \times 10^4 \text{ L mol}^{-1} \text{ s}^{-1}$

(4) $8.0 \times 10^{-4} \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$

144. In which one of the following processes the reactants and catalyst exist in three different states?

(1) Haber's process

- (2) Ostwald's process
- (3) Hydrogenation of Vegetable oil
- (4) Contact process

145. What is the slag formed in the extraction of iron?

- (1) CaO
- (2) CaSiO_3
- (3) MgSiO_3
- (4) SiO_2

146. Calcium phosphide reacts with water to form $\text{Ca}(\text{OH})_2$ and X.

When X is passed into CuSO_4 solution, Y and H_2SO_4 are formed. What is Y?

- (1) $[\text{Cu}(\text{PH}_3)_4]^{2+}$
- (2) $[\text{Cu}(\text{PH}_3)_6]^{2+}$
- (3) Cu_3P_2
- (4) CuHPO_4

147. Identify the statements which are not correct.

- i. ZnO , PbO , Sb_2O_3 are neutral oxides.
- ii. CO and NO are amphoteric oxides.
- iii. CrO_3 , Mn_2O_7 , V_2O_5 are basic oxides

- (1) i, ii
- (2) i, iii
- (3) ii, iii
- (4) i, ii, iii

148. Which one of the following liberates oxygen immediately when passed into water?

- (1) F_2
- (2) Cl_2
- (3) Br_2
- (4) I_2

149. Assertion (A) : CuI_2 cannot be prepared by the reaction of Cu^{2+} (aq) with I^- (aq).

Reason (R) : Aqueous Cu^{2+} solution is blue in colour

The correct answer is

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct and (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

150. A coordination compound is made of Co^{3+} , NH_3 and Cl^- . 0.1 M solution of this complex when treated with excess silver nitrate gave no precipitate. The formula of the complex and secondary valency of metal are respectively.

- (1) $[\text{Co}(\text{NH}_3)_3 \text{Cl}_3]$, 6
- (2) $[\text{Co}(\text{NH}_3)_5 \text{Cl}] \text{Cl}_2$, 6
- (3) $[\text{Co}(\text{NH}_3)_3 \text{Cl}_3]$, 3
- (4) $[\text{Co}(\text{NH}_3)_4 \text{Cl}_2] \text{Cl}$, 6

151. Nylon 6, 6 is a condensation polymer of two monomers X and Y. The number of $-\text{CH}_2-$ groups in X and Y are respectively

- (1) 6, 4

(2) 6, 6

(3) 5, 6

(4) 6, 2

152. Reducing saccharides among the following are

Sucrose	Ribose	Maltose	Lactose	Cellulose
1	2	3	4	5

(1) 2, 4, 5

(2) 1, 3, 4

(3) 2, 3, 5

(4) 2, 3, 4

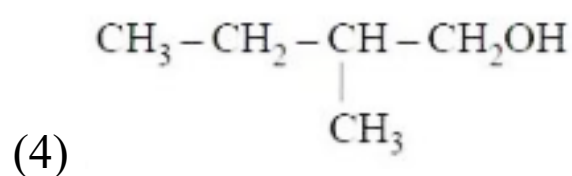
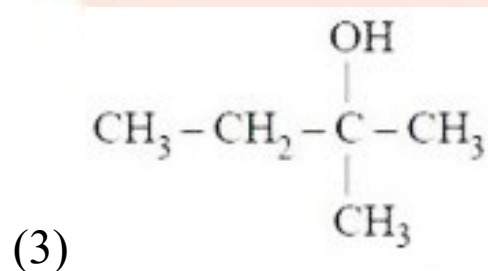
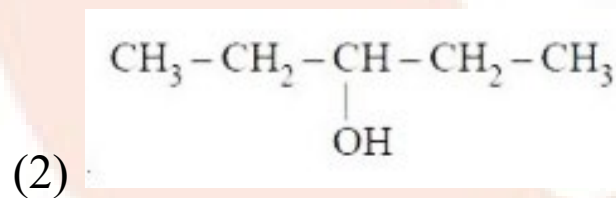
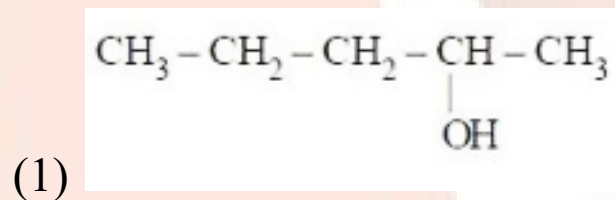
153. Example of antihistamine (X) and cationic detergent (Y) are

X	Y
(1) Dimetane	Cetyl trimethyl ammonium bromide
(2) Nardil	Cetyl trimethyl ammonium bromide
(3) Dimetane	Sodium lauryl sulphate
(4) Nardil	Sodium lauryl sulphate

154. S_N2 reaction involving inversion of configuration takes place with an optically active compound Z. The compound Z is

- (1) $\text{CH}_3\text{CH}_2\text{X}$
- (2) $(\text{CH}_3)_2\text{CHX}$
- (3) $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{X}$
- (4) $(\text{CH}_3)_3\text{CX}$

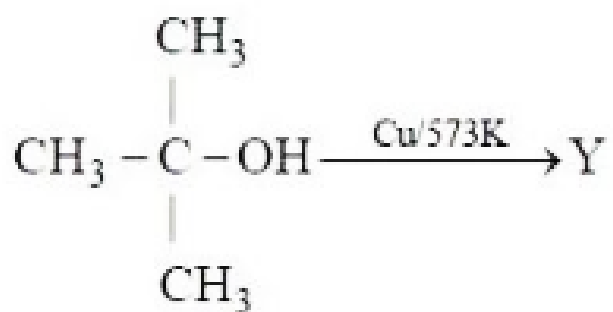
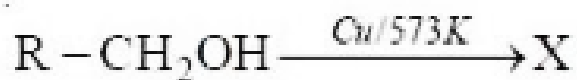
155. Ethyl magnesium bromide reacts with acetone to give X. On hydrolysis X forms



156. Identify the correct set from the following.

Compound	pK_a
(1) m – nitrophenol,	10.2
(2) o – nitrophenol,	7.2
(3) m – nitrophenol,	7.2
(4) o – nitrophenol,	7.2

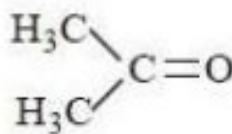
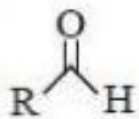
157. What are X and Y in the following reactions?

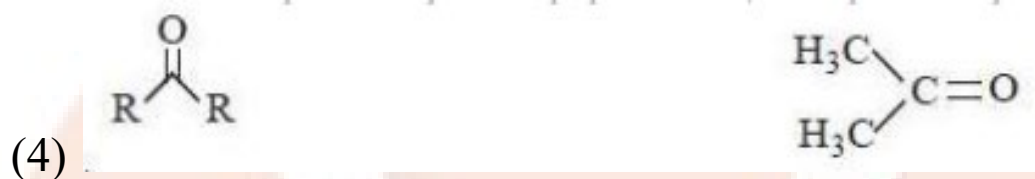
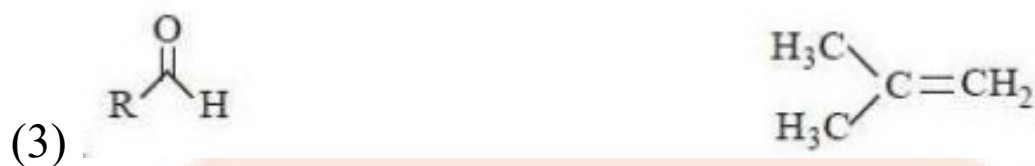


X

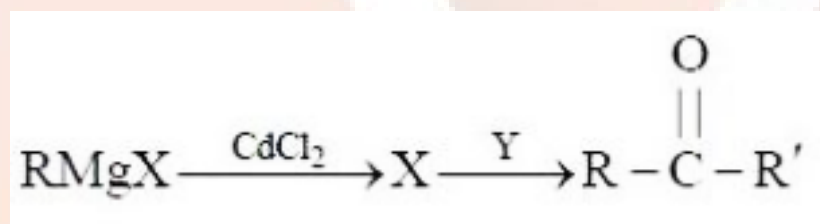
Y

(1)





158. X and Y in the following reaction sequence are

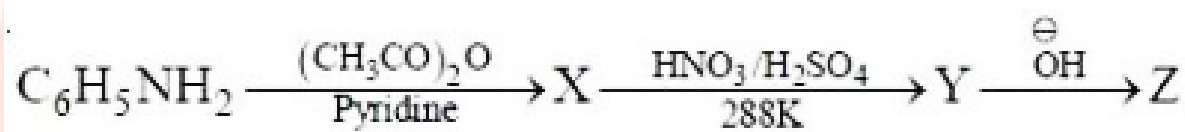


- | <u>X</u> | <u>Y</u> |
|---------------------------------|--------------------------|
| (1) RCdCl | $\text{R}'\text{COCl}$ |
| (2) R_2Cd | $(\text{R}'\text{CO})_2$ |
| (3) R_2Cd | $\text{R}'\text{COCl}$ |
| (4) $\text{R}_2\text{MgCdCl}_2$ | RCOOR' |

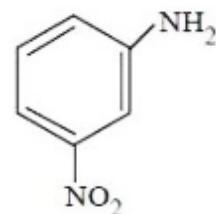
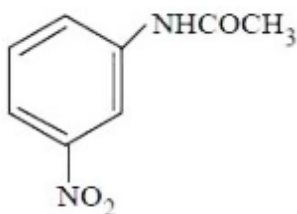
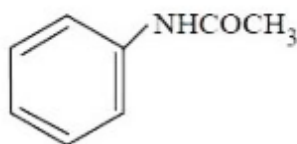
159. Order of acidity of benzoic acid (I), 4-methoxybenzoic acid (II), acetic acid (III) and 4-nitrobenzoic acid (IV) is

- (1) $\text{IV} > \text{I} > \text{II} > \text{III}$
 (2) $\text{I} > \text{II} > \text{IV} > \text{III}$
 (3) $\text{III} > \text{I} > \text{II} > \text{IV}$
 (4) $\text{II} > \text{I} > \text{IV} > \text{III}$

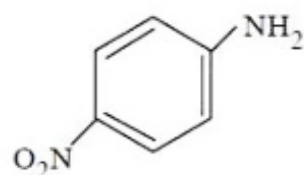
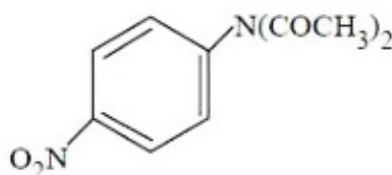
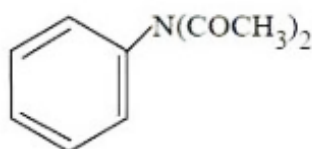
160. What are the structures of X, Y and Z in the following reaction sequence?

XYZ

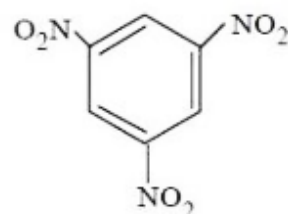
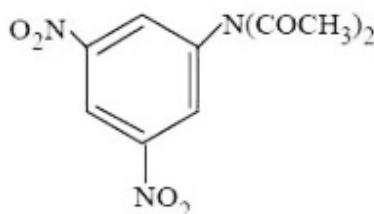
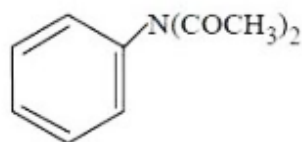
(1)



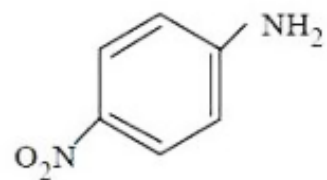
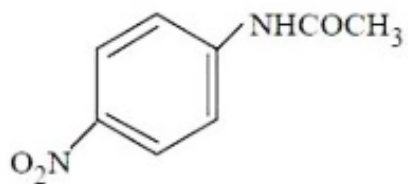
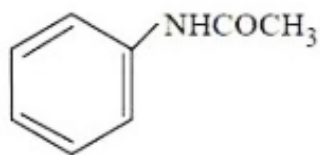
(2)



(3)



(4)



AP EAMCET-2018-Engineering-23 Apr-Shift-2
Answer Key

1	1	33	1	65	2	97	2	129	4
2	4	34	2	66	2	98	1	130	2
3	3	35	3	67	3	99	3	131	4
4	4	36	3	68	3	100	3	132	1
5	2	37	4	69	2	101	3	133	2
6	2	38	2	70	2	102	4	134	2
7	3	39	1	71	2	103	1	135	2
8	3	40	2	72	4	104	4	136	2
9	1	41	2	73	1	105	*	137	2
10	3	42	2	74	3	106	3	138	4
11	1	43	2	75	3	107	1	139	4
12	1	44	2	76	4	108	1	140	1
13	2	45	1	77	4	109	2	141	3
14	3	46	1	78	1	110	2	142	2
15	4	47	1	79	2	111	3	143	2
16	3	48	3	80	4	112	1	144	3
17	3	49	1	81	2	113	2	145	2

18	2	50	4	82	1	114	3	146	3
19	4	51	2	83	4	115	2	147	4
20	4	52	1	84	4	116	4	148	1
21	1	53	3	85	4	117	4	149	2
22	3	54	3	86	1	118	2	150	1
23	4	55	4	87	2	119	4	151	1
24	1	56	2	88	2	120	3	152	4
25	4	57	2	89	1	121	3	153	1
26	1	58	3	90	2	122	2	154	3
27	4	59	3	91	4	123	3	155	3
28	3	60	2	92	3	124	1	156	4
29	3	61	1	93	4	125	4	157	3
30	2	62	2	94	2	126	3	158	3
31	2	63	2	95	4	127	3	159	1
32	1	64	2	96	2	128	1	160	4

AP EAMCET-2018-Engineering-23 Apr-Shift-2
Solutions – Mathematics

1. Consider the expression,

$$g(x) = x^2 + x - 2$$

And,

$$\frac{1}{2}(g \circ f)(x) = 2x^2 - 5x + 2$$

Then,

$$\frac{1}{2}(g \circ f)(x) = 4x^2 - 10x + 4$$

And,

$$(f(x)^2) + f(x) - 2 = 4x^2 - 10x + 4$$

For the function to be linear $f(x) = ax + b$ so,

$$(ax + b)^2 + (ax + b) - 2 = 4x^2 - 10x + 4$$

$$a^2x^2 + (2ab + a)x + (b^2 + b - 2) = 4x^2 - 10x + 4$$

By the comparison of coefficients,

$$a^2 = 4$$

And,

$$2ab + a = -10$$

And,

$$b^2 + b - 2 = 4$$

So,

$$a = \pm 2$$

Then,

$$b = \begin{cases} -3; & \text{if } a = 2 \\ 2; & \text{if } a = -2 \end{cases}$$

These values satisfies above relations, thus,

$$f(x) = 2x - 3 \text{ or } -2x + 2$$

2. Consider the quadratic equation

$$x^2 + 2x + 2 = (x+1)^2 + 1 \in [1, \infty), \forall x \in R$$

Then,

$$\frac{1}{(x+1)^2 + 1} \in (0, 1]$$

For $f(x) = \frac{1}{x^2 + 2x + 2}$, $\forall x \in R$ is subjective then set,

$$A = (0, 1]$$

3. Consider the expression,

$$2 \cdot 4^{(2n+1)} + 3^{2n+1} = P(n)$$

$$P(n) = 8(4^{2n}) + 3(3^{3n})$$

For $n = 1$,

$$\begin{aligned} P(n=1) &= 128 + 81 \\ &= 209 \end{aligned}$$

Above is divisible by 19 and 11.

Similarly,

$$\begin{aligned} P(n=2) &= 2048 + 2187 \\ &= 4235 \end{aligned}$$

Above is divisible by 11 but not by 19.

So, the given expression is divisible by,

$$k = 11, n \in N$$

4. Consider the matrix,

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

The characteristic equation is given by,

$$A^2 + I = 0$$

So,

$$A^3 = -A$$

$$A^4 = I$$

Now by analyzing,

$$A^2 + I = A(A^2 - I)$$

is incorrect.

5. For non trivial solutions,

$$\begin{vmatrix} 1 & -a & -a \\ -b & 1 & -b \\ -c & -c & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} -\frac{1}{a} & 1 & -a \\ 1 & -\frac{1}{b} & 1 \\ 1 & 1 & -\frac{1}{c} \end{vmatrix} = 0$$

Apply the row operation $R_2 - R_1$ and $R_3 - R_1$

$$\begin{vmatrix} -\frac{1}{a} & 1 & 1 \\ 1+\frac{1}{a} & -\frac{1}{b}-1 & 0 \\ 1+\frac{1}{a} & 0 & -\frac{1}{c}-1 \end{vmatrix} = 0$$

$$-\frac{1}{a}\left(\frac{1}{b}+1\right)\left(\frac{1}{c}+1\right)+\left(1+\frac{1}{a}\right)\left(1+\frac{1}{c}\right)+\left(\frac{1}{b}+1\right)\left(1+\frac{1}{a}\right)=0$$

$$\frac{(1+a)(1+c)}{ac} + \frac{(1+a)(1+b)}{ab} = \frac{(1+b)(1+c)}{abc}$$

$$\frac{b}{1+b} + \frac{c}{1+c} = \frac{1}{1+a} - 1 + 1$$

Further simplify the above equation,

$$\frac{a}{1+a} + \frac{b}{1+b} + \frac{c}{1+c} = 1$$

6. Consider the matrix,

$$A = \begin{bmatrix} 1 & -2 & 3 & -4 \\ 2 & 9 & 4 & 5 \\ 4 & 5 & 10 & -3 \\ 1 & 11 & -1 & 9 \end{bmatrix}$$

$$R_3 = (R_2 + 2R_1) \text{ and } R_4 - (R_2 - R_1)$$

$$A = \begin{bmatrix} 1 & -2 & 3 & -4 \\ 2 & 9 & 4 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \end{bmatrix}$$

So, rank of A is 3.

7. Let

$$Z = x + iy$$

So,

$$|z|^2 + 1 = |z^2 - 1|$$

then

$$x^2 + y^2 + 1 = \sqrt{(x^2 - y^2 - 1) + 4x^2y^2}$$

$$(x^2 + y^2 + 1)^2 = (x^2 - y^2 - 1)^2 + 4x^2y^2$$

$$(x^2 + y^2 + 1)^2 - (x^2 - y^2 - 1)^2 = 4x^2y^2$$

$$[(x^2 + y^2 + 1) - (x^2 - y^2 - 1)] = 4x^2y^2$$

Then,

$$2x^2(2y^2 + 2) = 4x^2y^2$$

$$x^2y^2 + 2x^2 = x^2y^2$$

$$x^2 = 0$$

$$x = 0$$

8. Let,

$$z = x + iy$$

So,

$$\frac{z-i}{z-1} = \frac{x+i(y-1)}{(x-1)^2 + iy}$$

$$\operatorname{Re}\left(\frac{z-i}{z-1}\right) = \frac{x(x-1) + y(y-1)}{(x-1)^2 + y^2}$$

Since $\frac{z-i}{z+1}$ is purely imaginary so,

$$\operatorname{Re}\left(\frac{z-i}{z+1}\right) = 0$$

$$x^2 + y^2 - x - y = 0$$

It is a circle with center $\left(\frac{1}{2}, \frac{1}{2}\right)$ and radius $\frac{1}{\sqrt{2}}$ excluding the points $(1, 0)$ and $(0, 1)$.

9. Consider the expression,

$$\alpha = \frac{a+ib}{1-c}$$

$$|\alpha|^2 = \frac{a^2 + b^2}{(1-c)^2} \quad \dots\dots (I)$$

And,

$$a^2 + b^2 + c^2 = c$$

$$a^2 + b^2 = c(1 - c) \quad \text{..... (II)}$$

From equation (I) and (II),

$$\begin{aligned} |\alpha|^2 &= \frac{c}{1 - c} \\ &= \frac{1 + |\alpha|^2}{1} \\ &= \frac{1}{1 - c} \\ 1 - c &= \frac{1}{1 + |\alpha|^2} \quad \text{..... (III)} \end{aligned}$$

From equation (I) and (III),

$$\begin{aligned} |\alpha|^2 &= \frac{a^2 + b^2}{\left(\frac{1}{1 + |\alpha|^2}\right)^2} \\ a^2 + b^2 &= \frac{|\alpha|^2}{(1 + |\alpha|^2)^2} \end{aligned}$$

10. Consider the expression,

$$(1 + \sin \theta + i \cos \theta)^n + (1 + \sin \theta + i \cos \theta)^n$$

This implies,

$$\begin{aligned}
& \left[1 + \cos\left(\frac{\pi}{2} - \theta\right) + i \sin\left(\frac{\pi}{2} - \theta\right) \right]^n + \left[1 + \cos\left(\frac{\pi}{2} - \theta\right) - i \sin\left(\frac{\pi}{2} - \theta\right) \right]^n \\
&= \left[2 \cos^2\left(\frac{\pi}{4} - \frac{\theta}{2}\right) + 2i \sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \right]^n + \left[2 \cos^2\left(\frac{\pi}{4} - \frac{\theta}{2}\right) - 2i \sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \right]^n \\
&= 2^n \cos^n\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \left\{ \left[\cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right) + i \sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \right]^n + \left[\cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right) - i \sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \right]^n \right\} \\
&= 2^{n+1} \cos^n\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \cos\left(\frac{n\pi}{4} - \frac{n\theta}{2}\right)
\end{aligned}$$

11. The equation formed by elimination of x is,

$$\left(\frac{ly+m}{y+l} \right)^2 - \alpha \left(\frac{ly+m}{y+l} \right) + \beta = 0$$

$$(l^2 y^2 + m^2 + 2lmy) - \alpha(ly+m)(y+l) + \beta(y+l)^2 = 0$$

$$(l^2 - \alpha l + \beta)y^2 + (2lm - \alpha l^2 - \alpha m + 2\beta l)y + (m^2 - \alpha ml + \beta l^2) = 0$$

Above equation and $x^2 + \alpha x + \beta = 0$ has same roots so,

$$\frac{l^2 - \alpha l + \beta}{1} = \frac{2lm - \alpha l^2 - \alpha m + 2\beta l}{\alpha}$$

$$= \frac{m^2 - \alpha ml + \beta l^2}{\beta}$$

$$\beta l^2 - \beta \alpha l + \beta^2 = m^2 - \alpha ml + \beta l^2$$

$$\beta^2 - \beta m + \beta(m - \alpha l) - m(m - \alpha l) = 0$$

Further simplify the above,

$$\beta(\beta - m) + (m - \alpha l)(\beta - m) = 0$$

$$\beta = m, \alpha l = m$$

12. Consider the equation,

$$\frac{x^2 + 2x + 1}{x^2 + 2x + 7} = y, \quad y \neq 1$$

Then,

$$(y - 1)x^2 + 2(y - 1)x + (7y - 1) = 0$$

Since $x \in R$ so,

$$D \geq 0$$

This gives,

$$\begin{aligned}
 4(y-1)^2 - 14(y-1)(7y-1) &\geq 0 \\
 (y-1)[y-1-7y+1] &\geq 0 \\
 y(y-1) &\leq 0 \\
 y &\in [0, 1]
 \end{aligned}$$

Thus,

$$y \in [0, 1]$$

13. Consider the expression,

$$[x]^2 - 7[x] + 12 \leq 0$$

Solve further,

$$\begin{aligned}
 [x]^2 - 4[x] - 3[x] + 12 &\leq 0 \\
 ([x-4])([x]-3) &\leq 0 \\
 [x] &\in [3, 4] \\
 x &\in [3, 5)
 \end{aligned}$$

14. Let the roots of the equation $\alpha, \beta, \gamma, \delta$ for the equation,

$$\begin{aligned}
 x^4 - 2ax^3 + 4bx^2 + 8ax + 16 &= 0 \\
 \alpha \times \beta \times \gamma \times \delta &= 16 \quad \text{..... (I)}
 \end{aligned}$$

For the reciprocal equation roots are $p\alpha, p\beta, p\gamma$ and $p\delta$

So product of the roots is 1.

$$p^4 \alpha \beta \gamma \delta = 1 \quad \dots\dots (II)$$

From equation (I) and (II),

$$|p| = \frac{1}{2}$$

15. The number of three digit numbers in which 9 appears at hundred place only is,

$$n = 1 \times 9 \times 9$$

$$n = 81$$

The number of three digits numbers in which 9 appears at ten or unit place only is,

$$N = (8 \times 1 \times 9) \times 2$$

$$N = 144$$

So, the required number is,

$$\begin{aligned} n + N &= 81 + 144 \\ &= 225 \end{aligned}$$

16. Solve the given equation,

$$\begin{aligned}
 {}^{34}C_{10} + 3({}^{34}C_9) + 3({}^{34}C_8) + {}^{34}C_7 &= \left(\begin{aligned} &({}^{34}C_{10} + {}^{34}C_9) + \\ &2({}^{34}C_9) + 2({}^{34}C_8) + ({}^{34}C_8 + {}^{34}C_7) \end{aligned} \right) \\
 &= {}^{35}C_{10} + 2({}^{35}C_9) + {}^{35}C_8 \\
 &= {}^{36}C_{10} + {}^{36}C_9 \\
 &= {}^{37}C_{10}
 \end{aligned}$$

17. Let,

$$\begin{aligned}
 S &= (1+x)^{100} + 2x(1+x)^{99} + 3x^2(1+x)^{98} + \dots + 101x^{100} \\
 \frac{x}{1+x}S &= x(1+x)^{99} + 2x^2(1+x)^{98} + \dots + 100x^{100} + 101\frac{x^{101}}{1+x} \\
 \frac{S}{1+x} &= (1+x)^{100} + x(1+x)^{99} + x^2(1+x)^{98} + \dots + x^{100} - 101\frac{x^{101}}{1+x} \\
 \frac{S}{1+x} &= \frac{(1+x)^{100} \left[\left(\frac{x}{1+x} \right)^{101} - 1 \right]}{\frac{x}{1+x} - 1} - 101\frac{x^{101}}{1+x}
 \end{aligned}$$

Simplify the above,

$$S = (1+x)^{102} - x^{102} - 102x^{101}$$

So, the coefficient of x^{50} is $S = {}^{102}C_{50}$

18. Consider the expression,

$$(1+x)^n = \left(1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4 + \dots \right)$$

Comparison the given equation with above equation,

$$\frac{n(n-1)}{2!}x^2 = \frac{5}{2! \times 3} \text{ and } \frac{n(n-1)(n-2)}{3!}x^3 = \frac{5(7)}{3!(3^2)}$$

So,

$$x = -\frac{2}{3} \text{ and } n = -\frac{3}{2}$$

Therefore,

$$\left(1 - \frac{2}{3}\right)^{-\frac{3}{2}} = 1 + 1 + \alpha$$

$$3^{\frac{3}{2}} = 2 + \alpha$$

$$\alpha^2 + 4\alpha + 4 = 27$$

$$\alpha^2 + 4\alpha = 23$$

19. Consider the expression,

$$\frac{x^4 + 24x^2 + 28}{(x^2 + 1)^3} = \frac{A}{x^2 + 1} + \frac{B}{(x^2 + 1)^2} + \frac{C}{(x^2 - 1)^2}$$

$$x^4 + 24x^2 + 28 = A(x^2 + 1)^2 + B(x^2 + 1) + C(x^2 - 1)$$

Compare the coefficients,

$$A = 1,$$

And,

$$2A + B = 24$$

$$B = 22$$

And,

$$A + B + C = 28$$

$$C = 5$$

So,

$$A + C = 6$$

20. Consider the expression,

$$\begin{aligned}\tan \theta &= \frac{\cos 25^\circ + \sin 25^\circ}{\cos 25^\circ - \sin 25^\circ} \\ &= \frac{1 + \tan 25^\circ}{1 - \tan 25^\circ}\end{aligned}$$

The θ lies in third quadrant so,

$$\begin{aligned}\tan \theta &= \tan 70^\circ \\ &= \tan(180^\circ + 70^\circ) \\ &= \tan 250^\circ \\ \theta &= 250^\circ\end{aligned}$$

21. It is given that,

$$\cos A = \frac{7}{25} \text{ and } \frac{3\pi}{2} < A < 2\pi$$

Then,

$$\begin{aligned}\cos 2A &= 2\left(\frac{49}{625}\right) - 1 \\ &= \frac{98 - 625}{625} \\ &= -\frac{527}{625}\end{aligned}$$

And,

$$\cos \frac{A}{2} = -\frac{4}{5} \text{ and } \cos \frac{A}{4} = \frac{1}{\sqrt{10}}$$

So,

$$\cos \frac{A}{4} + \cos \frac{A}{2} - \cos 2A - \cos 2A = \frac{1}{\sqrt{10}} + \frac{7}{625}$$

22. Consider the expression,

$$\cos 3A + \cos 3B + \cos 3C + \cos 3\pi = 0$$

Solve above expression,

$$2 \cos \frac{3(A+B)}{2} \cos \frac{3(A-B)}{2} + \cos 3C - 1 = 0$$

$$-2 \sin \left(\frac{3C}{2} \right) \cos \frac{3(A-B)}{2} + 1 - 2 \sin^2 \left(\frac{3C}{2} \right) - 1 = 0$$

$$\sin \frac{3C}{2} \left[\cos \frac{3(A-B)}{2} - \cos \frac{3(A+B)}{2} \right] = 0$$

$$\sin \frac{3A}{2} \sin \frac{3B}{2} \sin \frac{3C}{2} = 0$$

It means either $A = \frac{2\pi}{3}$ or $B = \frac{2\pi}{3}$ or $C = \frac{2\pi}{3}$

So, the least value of the sum of angles is,

$$\pi - \frac{2\pi}{3} = \frac{\pi}{3}$$

23. Consider the expression,

$$\cos 2\theta \sec^4 \theta + \sec^2 \theta = 0, \theta \in \left(0, \frac{\pi}{2} \right)$$

Solve above equation,

$$(1 - \tan^2 \theta) \sec^2 \theta + \sec^2 \theta = 0$$

$$\sec^2 \theta [1 - \tan^2 \theta + 1] = 0$$

$$\tan^2 \theta = 2$$

$$\sin^2 \theta = \frac{2}{3}$$

24. Consider the expression,

$$r^4 + r^2 + 1 = (r^2 - r + 1)(r^2 + r + 1)$$

So,

$$\tan^{-1} \left(\frac{2r}{1 + (r^4 + r^2 + 1)} \right) = \tan^{-1} (r^2 + r + 1) - \tan^{-1} (r^2 - r + 1)$$

Solve further,

$$\begin{aligned} \sum_{r=1}^n \tan^{-1} \left(\frac{2r}{r^4 + r^2 + 2} \right) &= \sum_{r=1}^n \tan^{-1} (r^2 + r + 1) - \tan^{-1} (r^2 - r + 1) \\ &= \tan^{-1} (n^2 + n + 1) - \tan^{-1} (1) \\ &= \tan^{-1} \frac{n^2 + n}{1 + n^2 + n + 1} \\ &= \tan^{-1} \frac{n^2 + n}{n^2 + n + 2} \end{aligned}$$

So,

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \left(\frac{2r}{r^4 + r^2 + 2} \right) = \lim_{n \rightarrow \infty} \tan^{-1} \left(\frac{n^2 + n}{n^2 + n + 2} \right)$$

$$= \frac{\pi}{4}$$

25. Solve the given expression as,

$$\operatorname{sech}^2 \left(\operatorname{sech}^{-1} \left(\frac{\sqrt{3}}{2} \right) \right) + \operatorname{cosech}^2 \left(\operatorname{cosech}^{-1} (3) \right)$$

$$= \operatorname{sech}^2 \left(\operatorname{sech}^{-1} \left(\frac{\sqrt{3}}{2} \right) \right) + \operatorname{cosech}^2 \left(\operatorname{cosech}^{-1} (\sqrt{8}) \right)$$

$$= \frac{3}{4} + 8$$

$$= \frac{35}{4}$$

26. In triangle ABC,

$$a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$$

$$a \frac{s(s-c)}{ab} + c \frac{s(s-a)}{bc} = \frac{3b}{2}$$

$$\frac{s}{b} (2s - a - c) = \frac{3b}{2}$$

$$\frac{s}{b} (b) = \frac{3b}{2}$$

Further simplify the above,

$$\frac{a+b+c}{b} = 3$$

$$2b = a + c$$

27. Let,

$$\frac{s-a}{4} = \frac{s-b}{5} = \frac{s-c}{6} = k$$

As,

$$b + c - a = 8k$$

$$a + c - b = 10k$$

$$a + b - c = 12k$$

So,

$$a + b + c = 30$$

Hence,

$$s = 15k, a = 11k, b = 10k, c = 9k$$

The required solution is,

$$\begin{aligned}
 \sum \sin^2 \left(\frac{A}{2} \right) &= \frac{(s-a)(s-c)}{bc} + \frac{(s-c)(s-a)}{ca} + \frac{(s-a)(s-b)}{ab} \\
 &= \frac{30k^2}{90k^2} + \frac{24k^2}{99k^2} + \frac{20k^2}{110k^2} \\
 &= \frac{1}{3} + \frac{8}{33} + \frac{2}{11} \\
 &= \frac{25}{33}
 \end{aligned}$$

28. In the triangle IBC, $\angle BIC = \frac{\pi}{2} - \frac{A}{2}$

Let circumference of above be C_1 . Then $\angle BC_1C = \frac{\pi}{2} + \frac{A}{2}$

So,

$$\begin{aligned}
 \frac{a}{P_1} &= \sin \left(\frac{\pi}{2} + \frac{A}{2} \right) \\
 &= \cos \frac{A}{2}
 \end{aligned}$$

$$P_1 = \frac{a}{2 \cos \frac{A}{2}}$$

Similarly,

$$P_2 = \frac{b}{2 \cos \frac{B}{2}}, P_2 = \frac{c}{2 \cos \frac{C}{2}}$$

So,

$$\begin{aligned} P_1 P_2 P_3 &= \frac{abc}{8 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}} \\ &= \frac{abc \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}}{\sin A \sin B \sin C} \\ &= (2R)(2R)(2R) \frac{r}{4R} \\ &= 2R^2 r \end{aligned}$$

29. If the resultant of two vectors is unit vector then angle between them is $\frac{2\pi}{3}$, so,

$$\begin{aligned} |\mathbf{a} - \mathbf{b}|^2 &= |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}|\cos \frac{2\pi}{3} \\ &= 1 + 1 - 2(1)(1)\left(-\frac{1}{2}\right) \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

30. The equation of plane joining points $(\hat{i} + \hat{j})$ and $(3\hat{i} + \hat{j} - \hat{k})$ in

Cartesian form is,

$$\frac{x-1}{2} = \frac{y-1}{0} = \frac{z}{-1} = r$$

Then, point on above line is $p(2r+1, 1, -r)$.

Now the equation of the plane passing through $(2\hat{i} + 4\hat{j})$ and parallel to the vector $(3\hat{j} + 5\hat{k})$ and $(3\hat{i} - \hat{k})$ in Cartesian form is,

$$\begin{vmatrix} x-2 & y-4 & z \\ 0 & 3 & 5 \\ 3 & 0 & -1 \end{vmatrix} = 0$$

$$-3(x-2) + 15(y-4) - 9z = 0$$

$$3x - 15y + 9z + 54 = 0$$

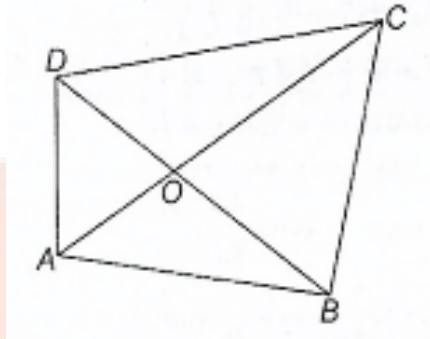
Let the point $P(2r+1, 1, -r)$ on the above plane itself then,

$$r = 14$$

So, the position vector of point P is,

$$\mathbf{OP} = 29\hat{i} + \hat{j} - 14\hat{k}$$

31. Let A, B, C, D be the vertices of quadrilateral as shown below.



so,

$$|\mathbf{AC}|^2 + |\mathbf{BD}|^2 = |\mathbf{AB}|^2 + |\mathbf{CD}|^2 + 2\mathbf{BC} \cdot \mathbf{AD} \quad (\text{I})$$

And,

$$|\mathbf{AC}|^2 + |\mathbf{BD}|^2 = |\mathbf{AD}|^2 + |\mathbf{BC}|^2 + 2\mathbf{AB} \cdot \mathbf{DC} \quad (\text{II})$$

From equation (I) and (II),

$$\mathbf{BC} \cdot \mathbf{AD} = \mathbf{AB} \cdot \mathbf{DC}$$

So,

$$(\mathbf{OC} - \mathbf{OB}) \cdot (\mathbf{OD} - \mathbf{OA}) = (\mathbf{OB} - \mathbf{OA}) \cdot (\mathbf{OC} - \mathbf{OD})$$

$$\begin{pmatrix} \mathbf{OC} \cdot \mathbf{OD} - \mathbf{OC} \cdot \mathbf{OA} \\ -\mathbf{OB} \cdot \mathbf{OD} + \mathbf{OB} \cdot \mathbf{OA} \end{pmatrix} = \begin{pmatrix} \mathbf{OB} \cdot \mathbf{OC} - \mathbf{OB} \cdot \mathbf{OD} \\ -\mathbf{OA} \cdot \mathbf{OC} + \mathbf{OA} \cdot \mathbf{OD} \end{pmatrix}$$

$$\mathbf{OC} \cdot \mathbf{OD} + \mathbf{OB} \cdot \mathbf{OA} = \mathbf{OB} \cdot \mathbf{OC} + \mathbf{OA} \cdot \mathbf{OD}$$

$$\mathbf{OC} \cdot (\mathbf{OD} - \mathbf{OB}) + \mathbf{OA} \cdot (\mathbf{OB} - \mathbf{OD}) = 0$$

Solve further,

$$\mathbf{OC} \cdot \mathbf{BD} + \mathbf{OA} \cdot \mathbf{DB} = 0$$

$$\mathbf{BD} \cdot (\mathbf{OC} - \mathbf{OA}) = 0$$

$$\mathbf{BD} \cdot \mathbf{AC} = 0$$

32. Any vector $\mathbf{r}$ can be written in linear combination of two non parallel vectors $\mathbf{b}$ and $\mathbf{c}$ as,

$$\mathbf{r} = \left(\frac{\mathbf{r} \cdot \mathbf{b}}{|\mathbf{b}|^2} \right) \mathbf{b} + \left(\frac{\mathbf{r} \cdot \mathbf{c}}{|\mathbf{c}|^2} \right) \mathbf{c} + \left(\frac{\mathbf{r} \cdot (\mathbf{b} \times \mathbf{c})}{|\mathbf{b} \times \mathbf{c}|^2} \right) (\mathbf{b} \times \mathbf{c})$$

$$\mathbf{r} = \mathbf{a}$$

$$|\mathbf{r}| = |\mathbf{a}|$$

$$= \sqrt{1+4+9}$$

Further simplify the above,

$$|\mathbf{r}| = \sqrt{14}$$

33. Solve the given expression as,

$$[\mathbf{abc}]^2 = \begin{vmatrix} \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{c} \cdot \mathbf{b} & \mathbf{c} \cdot \mathbf{c} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 4 & 2 \\ 2 & 2 & 9 \end{vmatrix}$$

$$= 15$$

34. Consider the given data,

$$(\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) = [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] \mathbf{c}$$

$$(\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b}) = [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] \mathbf{a}$$

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{b} \times \mathbf{c}) = [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] \mathbf{b}$$

And,

$$[\mathbf{b} + \mathbf{c} \ \mathbf{c} + \mathbf{a} \ \mathbf{a} + \mathbf{b}] = 2[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$$

$$[\mathbf{b} \times \mathbf{c} \ \mathbf{c} \times \mathbf{a} \ \mathbf{a} \times \mathbf{b}] = [\mathbf{a} \ \mathbf{b} \ \mathbf{c}]^2$$

Then,

$$\frac{(\mathbf{b} \times \mathbf{c})(\mathbf{c} \times \mathbf{a})(\mathbf{c} \times \mathbf{a})(\mathbf{a} \times \mathbf{b})(\mathbf{a} \times \mathbf{b}) \times (\mathbf{b} \times \mathbf{c})}{[\mathbf{b} \times \mathbf{c} \ \mathbf{c} \times \mathbf{a} \ \mathbf{a} \times \mathbf{b}][\mathbf{b} \times \mathbf{c} \ \mathbf{c} \times \mathbf{a} \ \mathbf{a} \times \mathbf{b}]} = \frac{[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] \mathbf{c} [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] \mathbf{a} [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] \mathbf{b}}{2[\mathbf{a} \ \mathbf{b} \ \mathbf{c}][\mathbf{a} \ \mathbf{b} \ \mathbf{c}]^2}$$

$$\frac{[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]^3 [\mathbf{a} \ \mathbf{b} \ \mathbf{c}]}{3[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]^3} = \frac{[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]}{2}$$

$$= 1$$

35. Consider the given data,

505, 510, 515, 520, ... 595

The mean of above is,

$$\bar{x} = 500$$

So,

$$\begin{aligned}\sum_{i=1}^{19} (x_i - \bar{x})^2 &= 2(45^2 + 40^2 + 35^2 + \dots + 5^2) \\ &= 2 \times 5^2 [1^2 + 2^2 + 3^2 + \dots + 9^2]\end{aligned}$$

So, the standard deviation is,

$$\begin{aligned}\sqrt{\frac{\sum_{i=1}^{19} (x_i - \bar{x})^2}{n}} &= \sqrt{\frac{2 \times 5^2 \times \frac{9 \times 10 \times 19}{6}}{19}} \\ &= 5\sqrt{30}\end{aligned}$$

36. The mean of the given data (x_i) is,

$$\begin{aligned}\frac{[(4 \times 3) + (8 \times 5) + (11 \times 9) + (17 \times 5) + \\ (20 \times 4) + (24 \times 3) + (32 \times 1)]}{3 + 5 + 9 + 5 + 4 + 3 + 1} &= \frac{420}{30} \\ &= 14\end{aligned}$$

So, the variance is,

$$k \frac{10^2 + 6^2 + 3^2 + 3^2 + 6^2 + 10^2 + 18^2}{30} = 45.8$$

The mean of the given data (y_i) is,

$$\begin{aligned}\frac{[(10 \times 3) + (18 \times 5) + (24 \times 9) + (36 \times 5) + \\ (42 \times 4) + (50 \times 3) + (66 \times 1)]}{3 + 5 + 9 + 5 + 4 + 3 + 1} &= \frac{900}{30} \\ &= 30\end{aligned}$$

So, the variance is,

$$\begin{aligned} k \frac{20^2 + 12^2 + 6^2 + 6^2 + 12^2 + 20^2 + 36^2}{30} &= 4 \times k \frac{\left(10^2 + 6^2 + 3^2 + 3^2 + 6^2 + 10^2 + 18^2 \right)}{30} \\ &= 4 \times 45.8 \\ &= 183.2 \end{aligned}$$

37. Let the probability of getting number is $P(i)$ then,

$$P(i) = Ki$$

Then,

$$P(1) = K, P(2) = 2K, P(3) = 3K$$

$$P(4) = 4K, P(5) = 5K, P(6) = 6K$$

This implies,

$$\sum P(i) = 1$$

$$21K = 1$$

$$K = \frac{1}{21}$$

So, the probability for getting odd number is,

$$\begin{aligned}P(1) + P(3) + P(5) &= 9K \\&= \frac{9}{21} \\&= \frac{3}{7}\end{aligned}$$

38. The required probability is given by,

$$\begin{aligned}&= P(A)P(\bar{B})P(\bar{C}) + P(\bar{A})P(B)P(\bar{C}) + P(\bar{A})P(\bar{B})P(C) \\&= \frac{1}{2}\left(\frac{2}{3}\right)\left(\frac{3}{4}\right) + \frac{1}{2}\left(\frac{1}{3}\right)\left(\frac{3}{4}\right) + \frac{1}{2}\left(\frac{2}{3}\right)\left(\frac{1}{4}\right) \\&= \frac{1}{4} + \frac{1}{8} + \frac{1}{12} \\&= \frac{11}{24}\end{aligned}$$

39. Let event

E_1 = he guesses the answer, E_2 = he knows the answer and

A = he answers correctly

Then,

$$P(E_2) = \frac{9}{10}, P(E_1) = \frac{1}{10}, P(A|E_2) = 1, P(A|E_1) = \frac{1}{4}$$

So,

$$\begin{aligned}P(E_1|A) &= \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)} \\&= \frac{\frac{1}{10} \times \frac{1}{4}}{\left(\frac{1}{10} \times \frac{1}{4}\right) + \left(\frac{9}{10} \times 1\right)} \\&= \frac{1}{1 + 36} \\&= \frac{1}{37}\end{aligned}$$

40. Since,

$$\sum P(X = x_i) = 1$$

Then,

$$0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$$

$$9K + 10K^2 = 0$$

$$K = \frac{1}{10}$$

So,

$$\begin{aligned}P(0 < K < 6) &= K + 2K + 2K + 3K + K^2 \\&= 8K + K^2 \\&= \frac{8}{10} + \frac{1}{100} \\&= \frac{81}{100} \\&= \left(\frac{9}{10}\right)^2\end{aligned}$$

41. The probability of getting at least one head is,

$$\begin{aligned}1 - (\text{Probability of getting no head}) &= 1 - {}^nC_n \left(\frac{1}{2}\right)^n \\&= 1 - \left(\frac{1}{2}\right)^n > 0.8\end{aligned}$$

Solve further,

$$\begin{aligned}\left(\frac{1}{2}\right)^n &> \frac{1}{4} \\2^n &> 5\end{aligned}$$

So, the least value of n is 3.

42. Let point $P(x, y)$ so,

$$\sqrt{(x-1)^2 + (y+1)^2} + \sqrt{(x+1)^2 + (y-1)^2} = 4$$

$$(x-1)^2 + (y+1)^2 + (x+1)^2 + (y-1)^2 = 16$$

$$2(x^2 + y^2 + 2) + 2\sqrt{(x-1)^2 + (y+1)^2} \sqrt{(x+1)^2 + (y-1)^2} = 16$$

$$(x^2 + y^2 + 2) + \sqrt{(x-1)^2 + (y+1)^2} \sqrt{(x+1)^2 + (y-1)^2} = 8$$

Then,

$$[(x-1)^2 + (y+1)^2][[(x+1)^2 + (y-1)^2]] = (6 - x^2 - y^2)^2$$

$$[x^2 + y^2 - 2x + 2y + 2][x^2 + y^2 + 2x - 2y + 2] = (x^2 + y^2 - 6)^2$$

$$(x^2 + y^2 + 2)^2 - 4(x - y)^2 = (x^2 + y^2 - 6)^2$$

$$[(x^2 + y^2 + 2) - (x^2 + y^2 - 6)][(x^2 + y^2 + 2) + (x^2 + y^2 - 6)] = 4(x - y)^2$$

Further simplify the above,

$$8(2x^2 + 2y^2 - 4) = 4(x^2 + y^2 - 2xy)$$

$$4x^2 + 4y^2 - 8 = x^2 + y^2 - 2xy$$

$$3x^2 + 2xy + 3y^2 = 8$$

43. The given transformed equation is,

$$3x^2 - 6xy + 8y^2 = 8$$

Now,

$$x' = x \cos \frac{\pi}{4} - y \sin \frac{\pi}{4} = \frac{x - y}{\sqrt{2}}$$

$$y' = x \sin \frac{\pi}{4} + y \cos \frac{\pi}{4} = \frac{x + y}{\sqrt{2}}$$

The equation before transformation is,

$$3 \left(\frac{x - y}{\sqrt{2}} \right)^2 - 6 \left(\frac{x - y}{\sqrt{2}} \right) \left(\frac{x + y}{\sqrt{2}} \right) + \left(\frac{x + y}{\sqrt{2}} \right)^2 = 8$$

$$3(x^2 + y^2 - 2xy) - 6(x^2 - y^2) - 8(x^2 + y^2 + 2xy) = 16$$

$$5x^2 + 17y^2 + 10xy - 16 = 0$$

44. Let point $A(a, 0)$ and $B(0, b)$ so equation of variable line is,

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{(I)}$$

Since the equation (I) passes through point (α, β) so,

$$\frac{\alpha}{a} + \frac{\beta}{b} = 1 \quad \text{(II)}$$

Now the centroid of triangle OAB is $\left(\frac{a}{3}, \frac{b}{3} \right) = (h, k)$

So,

$$a = 3h \text{ and } b = 3k \quad \text{(III)}$$

From equation (II) and (III),

$$\frac{\alpha}{3h} + \frac{\beta}{3k} = 1$$

On taking locus of point (h, k) we get,

$$\beta x + \alpha y - 3xy = 0$$

45. Let the slope of the line inclined with angle 60° with the line L is n' so,

$$\begin{aligned}\tan 60^\circ &= \left| \frac{n-1}{1+n} \right| \\ &= \sqrt{3}\end{aligned}$$

This implies,

$$\begin{aligned}(n-1)^2 &= 3(n+1)^2 \\ 2n^2 + 8n + 2 &= 0\end{aligned}$$

The root of the above is the slope of required line. Thus the product of slopes is 1.

46. The required area is given by,

$$\begin{aligned}\left| \frac{(6+6)(6+6)}{2(-3)-3(2)} \right| &= \frac{12(12)}{12} \\ &= 12\end{aligned}$$

47. The equation of the line perpendicular to the line $ax + by - 1 = 0$ and passes through origin is,

$$bx - ay = 0 \quad (I)$$

Now, altitude passes through the intersection of the line $2x + 3y - 1 = 0$ and $x + 2y - 1 = 0$

So,

$$-b - a = 0$$

$$a + b = 0 \quad (II)$$

From above the line $ax + by - 1 = 0$ becomes,

$$x - y - \frac{1}{a} = 0 \quad (III)$$

Now, the altitude perpendicular to the line $2x + 3y - 1 = 0$ and passes through origin is,

$$3x = 2y$$

The point of intersection line (III) and $x + 2y - 1 = 0$ is,

$$\left(\frac{1}{3} \left(1 + \frac{2}{a} \right), \frac{1}{3} \left(1 - \frac{1}{a} \right) \right) \text{ satisfy the line,}$$

$$a = -8 \text{ and } b = 8$$

Thus, $(a, b) = (-8, 8)$

48. The given pair of straight lines is,

$$x^2 - 4xy + y^2 = 0$$

Then,

$$x^2 - 4xy + 4y^2 = 3y^2$$

$$(x - 2y)^2 - (\sqrt{3}y)^2 = 0$$

$$x - (2 + \sqrt{3})y = 0$$

$$\text{or } x - (2 - \sqrt{3})y = 0$$

So, the perpendicular distance of above from point $(1, -1)$ is,

$$d_1 = \frac{|1 + 2 + \sqrt{3}|}{\sqrt{1 + (2 + \sqrt{3})^2}}, d_2 = \frac{|1 + 2 - \sqrt{3}|}{\sqrt{1 + (2 - \sqrt{3})^2}}$$

So,

$$\begin{aligned} d_1 d_2 &= \frac{|3 + \sqrt{3}|}{\sqrt{8 + 4\sqrt{3}}} \times \frac{|3 - \sqrt{3}|}{\sqrt{8 - 4\sqrt{3}}} \\ &= \frac{9 - 3}{\sqrt{64 - 48}} \\ &= \frac{6}{4} \\ &= \frac{3}{2} \end{aligned}$$

49. The distance between parallel lines $3x - 4y - 10 = 0$ and $3x - 4y + 30 = 0$ is the length of diameter of required circle so radius is,

$$\frac{1}{2} \frac{40}{\sqrt{9+16}} = 4$$

And midpoint of the intersection of the line $3x - 4y - 10 = 0$ and $3x - 4y + 30 = 0$, $x + 2y = 0$ is the center of required circle to center is,

$$\left(\frac{2-6}{2}, \frac{-1+3}{2} \right) = (-2, 1)$$

Then the equation of required circle is,

$$x^2 + y^2 + 4x - 2y - 11 = 0$$

50. Let required point of intersection be (x_1, y_1) . Then the equation of chord of contact with respect to given circle is,

$$xx_1 + yy_1 - 2(x + x_1) - 3(y + y_1) + 4 = 0$$

$$(x_1 - 2)x + (y_1 - 3)y + (4 - 3y_1 - 2x_1) = 0$$

Let above represent the line,

$$AB = 4x + 4y - 11 = 0$$

Then,

$$\frac{x_1 - 2}{4} - \frac{y_1 - 3}{4} = \frac{4 - 3y_1 - 2x_1}{-11} = K$$

Then,

$$x_1 = 4K + 2$$

$$y_1 = 4K + 3$$

And,

$$2x_1 + 3y_1 - 4 = 11K$$

From above,

$$8K + 4 + 12K + 9 - 4 = 11K$$

$$9K + 9 = 0$$

$$K = -1$$

Thus,

$$(x_1, y_1) = (-2, -1)$$

51. Let the equation of required circle be,

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (I)$$

The above passes through point $(3, 2)$ so,

$$9 + 4 + 6g + 4f + c = 0$$

$$6g + 4f + c + 13 = 0$$

The equation (I) bisects the circumference of circle $x^2 + y^2 = 15$

So, the common chord passes through the center of the circle,

$$x^2 + y^2 = 15$$

So,

$$c + 15 = 0$$

$$c = -15$$

Since equation (I) cuts the circle,

$$x^2 + y^2 + 4x + 6y + 3 = 0$$

Orthogonally so,

$$4g + 6f = c + 3$$

From above equations,

$$g = 3, f = -4, c = -15$$

So, the equation of the required circle is,

$$x^2 + y^2 + 6x - 8y - 15 = 0$$

52. The equation of the two circles are,

$$x^2 + y^2 + 2x + 4y - 20 = 0$$

$$x^2 + y^2 + 6x - 8y + 10 = 0$$

This implies,

$$2g_1g_2 + 2f_1f_2 = 6 - 16$$

$$= -10$$

$$= c_1 + c_2$$

The circle intersects orthogonally so have a common tangent.

Now the equation of common chord is,

$$2x - 6y + 15 = 0$$

So, the length of common chord is,

$$\begin{aligned} 2\sqrt{25 - \frac{(-2+12+15)^2}{4+36}} &= 2\sqrt{25 - \frac{625}{40}} \\ &= 2\sqrt{\frac{1000 - 625}{40}} \\ &= \frac{5\sqrt{3}}{\sqrt{2}} \end{aligned}$$

53. The equation of the circle passing through intersection of circles,

$$x^2 + y^2 + 4x + 6y - 12 = 0 \quad \text{(I)}$$

And,

$$x^2 + y^2 - 6x - 4y - 12 = 0 \quad \text{(II)}$$

Is,

$$x^2 + y^2 + 4x + 6y - 12 + \lambda(x^2 + y^2 - 6x - 4y - 12) = 0$$

Then,

$$x^2 + y^2 + \frac{4-6\lambda}{1+\lambda}x + \frac{6-4\lambda}{1+\lambda}y - 12 = 0 \quad \text{(III)}$$

Since the another circle,

$$x^2 + y^2 - 4x + 2y + 8 = 0 \quad (IV)$$

Cut the circle (III) orthogonally then,

$$-2\left(\frac{4-6\lambda}{1+\lambda}\right) + \left(\frac{6-4\lambda}{1+\lambda}\right) = 8-12$$

$$-8 + 12\lambda + 6 - 4\lambda = -4 - 4\lambda$$

$$12\lambda = -2$$

$$\lambda = -\frac{1}{6}$$

So, required equation of circle is,

$$6(x^2 + y^2 + 4x + 6y - 12) - (x^2 + y^2 - 6x - 6y - 12) = 0$$

$$5x^2 + 5y^2 + 30x + 40y - 60 = 0$$

$$x^2 + y^2 + 6x + 8y - 12 = 0$$

54. Let the equation of parabola having horizontal axis vertex at (h, k) is,

$$(y - k)^2 = 4a(x - h) \quad (I)$$

The parabola in equation (I) passes through $(-2, 1), (1, 2)$ and $(-1, 3)$ so,

$$(k - 1)^2 = 4a(-2 - h)$$

$$k^2 - 2k + 1 = -8a - 4ah \quad (II)$$

Then,

$$(k - 2)^2 = 4a(1 - h)$$

$$k^2 - 4k + 4 = 4a - 4ah \quad (\text{III})$$

And,

$$(3 - k)^2 = 4a(-1 - h)$$

$$k^2 - 6k + 9 = -4a - 4ah \quad (\text{IV})$$

From equation (II), (III) and (IV),

$$4a = \frac{2}{5}$$

55. Let the equation of tangent to parabola $y^2 = x$ having slope m is,

$$y = mx + \frac{1}{4m} \quad (\text{I})$$

For the common tangent to the circle,

$$x^2 + y^2 - 6y + 4 = 0 \quad (\text{II})$$

The radius is,

$$\sqrt{9-4} = \frac{\left|3 - \frac{1}{4m}\right|}{\sqrt{1+m^2}}$$

$$5(1+m^2) = 9 + \frac{1}{16m^2} - \frac{3}{2m}$$

$$m = \frac{1}{2}$$

So, the equation of the common tangent is,

$$y = \frac{1}{2}x + \frac{1}{2}$$

$$x - 2y + 1 = 0$$

56. Let the equation of ellipse be,

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Now,

$$a - ae = 2$$

$$a\left(1 - \frac{3}{5}\right) = 2$$

$$a = 5$$

So, $b = 4$

Now vertex comparing the vertex then,

$$6 - h = 5$$

$$h = 1$$

And,

$$1 - k = 0$$

$$k = 1$$

So, the required ellipse is,

$$\frac{(x-1)^2}{25} + \frac{(y-1)^2}{16} = 1$$

57. The equation of the tangent at point $\left(4\cos 2\theta, \frac{16}{\sqrt{11}}\sin 2\theta\right)$ on the ellipse $16x^2 + 11y^2 = 256$ is,

$$16x(4\cos 2\theta) + 11y\left(\frac{16}{\sqrt{11}}\sin 2\theta\right) = 256$$

$$16x(4\cos 2\theta) + \sqrt{11}y\sin 2\theta = 16 \quad \text{(I)}$$

Since the equation (I) is the tangent to the circle,

$$x^2 + y^2 - 2x - 15 = 0$$

So,

$$\frac{|4\cos 2\theta - 16|}{\sqrt{16\cos^2 2\theta + 11\sin^2 2\theta}} = 4$$

$$\cos^2 2\theta - 8\cos 2\theta + 16 = 11 + 5\cos^2 2\theta$$

$$4\cos^2 2\theta + 8\cos 2\theta - 5 = 0$$

$$\cos 2\theta = \frac{1}{2}$$

Then,

$$2\theta = \frac{\pi}{3} \text{ or } \left(-\frac{\pi}{3}\right)$$

$$\theta = \pm \frac{\pi}{6}$$

58. The equation of given curve is,

$$\frac{x^2}{4a^2} - \frac{y^2}{9b^2} = 1 \quad \text{(I)}$$

Let at point $(2a \sec \theta, 3b \tan \theta)$ on the curve in equation (I),

So, equation of tangent at point is,

$$\frac{\frac{x}{2a}}{\sec \theta} + \frac{\frac{y}{3b}}{\tan \theta} = 1$$

As per the question,

$$2a = \sec \theta \text{ and } 3b = -\tan \theta$$

So,

$$4a^2 - 9b^2 = 1$$

Take locus of point (a, b) then,

$$4x^2 - 9y^2 = 1$$

59. Let the point $P(-9, 12, -15)$ divides the line joining $A(1, -2, 3)$ and $B(-4, 5, -6)$ in ratio $\lambda : 1$ then,

$$-9 = \frac{-4\lambda + 1}{\lambda + 1}, \lambda = -2$$

So, the harmonic conjugate of point P with respect to the line segment AB will divide the line segment AB internally in ratio 2:1, so, point will be,

$$\left(\frac{-8+1}{3}, \frac{10-2}{3}, \frac{-12+3}{3} \right) = \left(-\frac{7}{3}, \frac{8}{3}, -3 \right)$$

60. Let the direction cosines of the line perpendicular to both L_1 and L_2 are l, m, n then,

$$-2l - m + n = 0 \text{ and } 3l - 3m + 4n = 0$$

So,

$$\frac{l}{-4+3} = \frac{m}{5} = \frac{n}{3} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{1+25+9}}$$

$$l, m, n = \pm \frac{1}{\sqrt{35}}, \pm \frac{5}{\sqrt{35}}, \pm \frac{3}{\sqrt{35}}$$

61. Mid point of the line segment joining the points $P(3,2,4)$ and $Q(-1,0,-2)$ is $R(1,1,1)$ and direction of the line segment PQ is $4,2,6$ so, direction ratio of normal to the plane is,

$$\langle a, b, c \rangle = \langle 4, 2, 6 \rangle$$

So, equation of plane will be,

$$4x + 2y + 6z + d = 0$$

Since the above plane bisect the line segment joining PQ so,

$$d = -12$$

Therefore,

$$ac + bd = 0$$

62. Consider the expression,

$$\cos \left[\lim_{x \rightarrow \infty} \frac{2\pi|x| + \pi x}{|x| - 3x} + \lim_{x \rightarrow 0} \frac{\cos \left(\frac{\pi}{2} \cos^2 x \right)}{x^2} \right]$$

Simplify the above,

$$\begin{aligned} & \cos \left[\lim_{x \rightarrow \infty} \frac{2\pi x + \pi x}{x - 3x} + \lim_{x \rightarrow 0} \frac{\cos \left(\frac{\pi}{2} (1 - \sin^2 x) \right)}{x^2} \right] \\ &= \cos \left[\lim_{x \rightarrow \infty} \frac{3\pi x}{-2x} + \lim_{x \rightarrow 0} \left(\frac{\sin \left(\frac{\pi}{2} \sin^2 x \right)}{\left(\frac{\pi}{2} \sin^2 x \right)} \times \frac{\frac{\pi}{2} \sin^2 x}{x^2} \right) \right] \\ &= \cos \left[-\frac{3\pi}{2} + \frac{\pi}{2} \right] \\ &= \cos(-\pi) \\ &= -1 \end{aligned}$$

63. Consider the expression,

$$\begin{aligned}
 & \lim_{x \rightarrow \infty} \frac{\left[6^2 + 12^2 + 18^2 + \dots + (6n)^2\right]^2}{\left[5 + 10 + 15 + \dots + 5n\right]\left[2^3 + 4^3 + 6^3 + \dots + 8n^3\right]} \\
 &= \lim_{x \rightarrow \infty} \frac{6^4 \left[1^2 + 2^2 + 3^2 + \dots + (n)^2\right]^2}{5 \left[1 + 2 + 3 + \dots + n\right] 2^3 \left[1^3 + 2^3 + 3^3 + \dots + n^3\right]} \\
 &= \lim_{x \rightarrow \infty} \frac{6^4 \left[\frac{n(n+1)(2n+1)}{6}\right]^2}{5 \left(\frac{n(n+1)}{2}\right) \times 2^3 \times \left(\frac{n(n+1)}{2}\right)^2} \\
 &= \lim_{x \rightarrow \infty} \frac{6^4 \left[\frac{\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right)}{6}\right]^2}{5 \left(\frac{1 + \frac{1}{n}}{2}\right) \times 2^3 \times \left(\frac{1 + \frac{1}{n}}{2}\right)^2} \\
 &= \frac{6^4 \left(\frac{1}{6^2}\right) (2^2)}{5 \left(\frac{1}{2}\right) (2^3) \left(\frac{1}{2^2}\right)}
 \end{aligned}$$

Further simplify the above,

$$\frac{\left[6^2 + 12^2 + 18^2 + \dots + (6n)^2\right]^2}{[5 + 10 + 15 + \dots + 5n][2^3 + 4^3 + 6^3 + \dots + 8n^3]} = \frac{144}{5}$$

64. Consider the function,

$$f(x) = \begin{cases} [x], & \text{if } x < 2 \\ [x] - 1 & \text{if } x \geq 2 \end{cases}$$

At $x = 2$, $f(2) = 1$

So, LHL is,

$$\lim_{h \rightarrow 0} [2 - h] = 1$$

And RHL is,

$$\begin{aligned} \lim_{h \rightarrow 0} ([2 + h] - 1) &= 2 - 1 \\ &= 1 \end{aligned}$$

So, $f(x)$ is continuous at $x = 2$ and $[x]$ is continuous in $[n, n + 1)$, $\forall n \in \text{integer}$

So, given function is continuous in the interval $[1, 3)$.

65. Consider the function for $a > 0$,

$$f(x) = \left(\frac{a+x}{1+x} \right)^{a+1+2x}$$

$$f(0) = a^{a+1}$$

This implies,

$$\log f(x) = (a+1+2x) \log \left(\frac{a+x}{1+x} \right)$$

Differentiate w.r.t x ,

$$\frac{f'(x)}{f(x)} = 2 \log \left(\frac{a+x}{1+x} \right) + (a+1+2x) \left[\frac{1}{a+x} - \frac{1}{1+x} \right]$$

So,

$$f'(0) = f(0) \left[2 \log(a) + (a+1) \left(\frac{1}{a} - 1 \right) \right]$$

This implies,

$$f(0) = a^{a+1}$$

So,

$$f'(0) = a^{a+1} \left\{ \frac{1-a^2}{a} + 2 \log a \right\}$$

66. Consider the given expression,

$$y = \log_2(\log_2 x)$$
$$y = \frac{\log_e \left(\frac{\log_e x}{\log_2 x} \right)}{\log_{e^2}}$$

Differentiate w.r.t x ,

$$\frac{dy}{dx} = \frac{1}{\log_{e^2} \frac{\log_e x}{\log_2 x}} \times \frac{1}{x \log_e 2}$$
$$\frac{dy}{dx} = \frac{1}{x \log_e x \log_e 2}$$

67. Consider the expression,

$$y = x \log \left(\frac{x}{2-3x} \right), \text{ for } 0 < x < \frac{2}{3}$$

Differentiate w.r.t x ,

$$\frac{dy}{dx} = x \left(\frac{1}{x} - \frac{-3}{2-3x} \right) + \log \left(\frac{x}{2-3x} \right)$$
$$\frac{dy}{dx} = \frac{2}{2-3x} + \log \left(\frac{x}{2-3x} \right)$$

Again differentiate the above,

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{-2}{(2-3x)^2}(-3) + \left(\frac{1}{x} - \frac{-3}{2-3x}\right) \\ &= \frac{6}{(2-3x)^2} + \frac{2}{x(2-3x)}\end{aligned}$$

So,

$$\begin{aligned}\frac{d^2y}{dx^2}\left(\text{at } x = \frac{1}{2}\right) &= \frac{6}{\left(\frac{1}{2}\right)^2} + \frac{2}{\frac{1}{2}\left(\frac{1}{2}\right)} \\ &= 24 + 8 \\ &= 32\end{aligned}$$

68. Let,

$$y = f(x)$$

$$= x^{\frac{1}{4}}$$

And,

$$x = 16$$

$$\Delta x = 2$$

So,

$$\begin{aligned}\Delta y &= \left(\frac{dy}{dx} \right) \Delta x \\ &= \frac{1}{4} \left(x^{-\frac{3}{4}} \right) \Delta x\end{aligned}$$

And,

$$\begin{aligned}\Delta y &= \frac{1}{4} \left(\frac{1}{8} \right) (2) \\ &= \frac{1}{16}\end{aligned}$$

And,

$$\begin{aligned}y &= (16)^{\frac{1}{4}} \\ &= 2\end{aligned}$$

So,

$$\begin{aligned}y + \Delta y &= 2 + \frac{1}{16} \\ &= 2.0625\end{aligned}$$

69. Consider the function,

$$f(x) = 3x^4 - 2x^3 - 6x^2 + 6x + 4$$

So,

$$\begin{aligned}f'(x) &= 12x^3 - 6x^2 - 12x + 6 \\&= 6(x-1)(x+1)\left(x - \frac{1}{2}\right)\end{aligned}$$

For maxima and minima,

$$f'(x) = 0$$

$$x = 1, \frac{1}{2}$$

Now,

$$f(1) = 5 \text{ (minimum)}$$

$$f\left(\frac{1}{2}\right) = \frac{87}{16} \text{ (maximum)}$$

So,

$$f(1) + f\left(\frac{1}{2}\right) = \frac{167}{16}$$

70. Consider the function,

$$f(x) = \cos x - \sin 2x$$

Is a continuous function in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and

differentiable in interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Now according to Lagrange mean value theorem,

$$c \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Such that,

$$f'(c) = \frac{f\left(\frac{\pi}{2}\right) - f\left(-\frac{\pi}{2}\right)}{\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)}$$

This implies,

$$-\sin c - 2\cos(2c) = 0$$

$$\sin c + 2 - 4\sin^2 c = 0$$

$$4\sin^2 c - \sin c - 2 = 0$$

So,

$$\sin c = \frac{1 \pm \sqrt{1+32}}{8}$$

$$= \frac{1 \pm \sqrt{33}}{8}$$

$$c = \sin^{-1}\left(\frac{1 \pm \sqrt{33}}{8}\right)$$

71. Consider the given data,

$$\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt} \text{ at } r = 8 \text{ m}$$

So,

$$4 = 4\pi(64)\frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{64\pi} \text{ m/min}$$

Then,

$$s = 4\pi r^2$$

$$\begin{aligned}\frac{ds}{dt} &= 4\pi(2r)\frac{dr}{dt} \\ &= 64\pi\left(\frac{1}{64\pi}\right) \\ &= 1 \text{ m}^2/\text{min}\end{aligned}$$

72. Consider the expression,

$$\int \frac{x}{(x-1)^2(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+2)}$$

Now by partial fraction,

$$\frac{x}{(x-1)^2(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+2)}$$

$$x = A(x-1)(x+2) + B(x+2) + C(x-1)^2$$

Compare the coefficients,

$$A = \frac{2}{9}, B = \frac{1}{4}, C = -\frac{2}{9}$$

So,

$$\begin{aligned}\int \frac{x}{x^3 - 3x + 2} dx &= \frac{2}{9} \int \frac{dx}{x-1} + \frac{1}{3} \int \frac{dx}{(x-1)^2} - \frac{2}{9} \int \frac{dx}{x+2} \\ &= \frac{2}{9} \log|x-1| - \frac{1}{3} \left(\frac{1}{(x-1)} \right) - \frac{2}{9} \log|x+2| + c \\ &= -\frac{1}{3} \left(\frac{1}{(x-1)} \right) + \frac{2}{9} \log \left| \frac{x-1}{x+2} \right| + c\end{aligned}$$

73. Consider the integral,

$$\begin{aligned}I &= \int \frac{\sqrt{x^2 + 1} [\log(x^2 + 1) - 2 \log x]}{x^4} dx \\ &= \int \frac{1}{x^3} \sqrt{1 + \frac{1}{x^2}} \log \left(1 + \frac{1}{x^2} \right) dx\end{aligned}$$

Let,

$$1 + \frac{1}{x^2} = t^2$$

Then,

$$-2 \frac{dx}{x^3} = 2t dt$$

So,

$$\begin{aligned} I &= \int t^2 \log t^2 dt \\ &= -2 \int t^2 \log t dt \\ &= -2 \left[\frac{t^3}{3} \log t - \int \left(\frac{t^3}{3} \times \frac{1}{t} \right) dt \right] \\ &= -\frac{2}{3} t^3 \log t + \frac{2}{9} t^3 + c \end{aligned}$$

Further simplify the above,

$$\frac{1}{9} t^3 [2 - 3 \log t^2] + c = \frac{1}{9} \left(1 + \frac{1}{x^2} \right)^{\frac{3}{2}} \left[2 - 3 \log \left(1 + \frac{1}{x^2} \right) \right] + c$$

74. Consider the expression,

$$\begin{aligned} I &= \int \frac{dx}{\sin x + \sin 2x} \\ &= \int \frac{dx}{\sin x (1 + 2 \cos x)} \\ &= \int \frac{\sin x dx}{\sin^2 x (1 + 2 \cos x)} \\ &= \int \frac{\sin x dx}{(1 - \cos x)(1 + \cos x)(1 + 2 \cos x)} \end{aligned}$$

Let $\cos x = t$ then,

$$-\sin x dx = dt$$

So,

$$I = -\int \frac{dt}{(1-t)(1+t)(1+2t)}$$

By partial fraction,

$$\frac{1}{(1-t)(1+t)(1+2t)} = \frac{A}{1-t} + \frac{B}{1+t} + \frac{C}{1+2t}$$

$$1 = A(1+t)(1+2t) + B(1-t)(1+2t) + C(1+t)(1-t)$$

Then,

$$A = \frac{1}{6}, B = \frac{-1}{2}, C = \frac{4}{3}$$

So,

$$\begin{aligned} I &= -\frac{1}{6} \int \frac{dt}{1-t} + \frac{1}{2} \int \frac{dt}{1+t} - \frac{4}{3} \int \frac{dt}{1+2t} \\ &= \frac{1}{6} \log(1 - \cos x) + \frac{1}{2} \log(1 + \cos x) - \frac{2}{3} \log(1 + 2 \cos x) + c \end{aligned}$$

75. Consider the expression,

$$\begin{aligned} I &= \int (\sec^4 x + \tan^4 x) dx \\ &= \int (\sec^4 x + (\sec^2 x - 1)^2) dx \\ &= \int (2 \sec^4 x - 2 \sec^2 x + 1) dx \\ &= \int [2(1 + \tan^2 x) \sec^2 x - 2 \sec^2 x + 1] dx \end{aligned}$$

Then,

$$\begin{aligned}
 I &= \int (2 \tan^2 x \sec^2 x + 1) dx \\
 &= \frac{2}{3} \tan^3 x + x + c
 \end{aligned}$$

76. Consider the expression,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \sin^5 \left(\frac{\pi}{6n} \right) + \sin^5 \left(\frac{2\pi}{6n} \right) + \sin^5 \left(\frac{3\pi}{6n} \right) + \dots + \sin^5 \left(\frac{\pi}{2} \right) \right\}$$

Simplify the above,

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \sin^5 \left(\frac{\pi}{6n} \right) + \sin^5 \left(\frac{2\pi}{6n} \right) + \sin^5 \left(\frac{3\pi}{6n} \right) + \dots + \sin^5 \left(\frac{3n\pi}{6n} \right) \right\} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{3n} \sin^5 \left(\frac{r\pi}{6n} \right) \\
 &= \int_0^3 \sin^5 \left(\frac{\pi}{6} x \right) dx
 \end{aligned}$$

Let, $\frac{\pi}{6} x = t$

Then for upper limit, $x = 3, t = \frac{\pi}{2}$

And for lower limit $x = 0, t = 0$ and $dx = \frac{6}{\pi} dt$

So,

$$\begin{aligned}\int_0^3 \sin^5\left(\frac{\pi}{6}x\right)dt &= \frac{6}{\pi} \int_0^{\frac{\pi}{2}} \sin^5(t)dt \\ &= \frac{6}{\pi} \left(\frac{4(2)}{5(3)(1)} \right) \\ &= \frac{16}{5\pi}\end{aligned}$$

77. Consider the expression,

$$I = \int_0^{\frac{\pi}{2}} \log_e(\sin 2x) dx$$

Let $2x = t$

For lower limit, $x = 0, t = 0$

And for upper limit $x = \frac{\pi}{2}, t = \pi$ and $dx = \frac{1}{2} dt$

So,

$$\begin{aligned}I &= \frac{1}{2} \int_0^{\pi} \log_e \sin(t) dt \\ &= \int_0^{\frac{\pi}{2}} \log_e \sin(t) dt \\ &= -\frac{\pi}{2} \log_e 2\end{aligned}$$

78. The required area is given by,

$$\begin{aligned} 2 \int_0^2 \sqrt{8x} dx &= 4\sqrt{2} \left[\frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^2 \\ &= \frac{8\sqrt{2}}{3} (2\sqrt{2}) \\ &= \frac{32}{3} \end{aligned}$$

79. Consider the differential equation,

$$y = px + \sqrt{a^2 p^2 + b^2}$$

Then,

$$\sqrt{a^2 p^2 + b^2} = (y - px)$$

$$a^2 p^2 + b^2 = y^2 + p^2 x^2 - 2xyp$$

$$(x^2 - a^2) p^2 - 2xyp + y^2 - b^2 = 0$$

Thus, order and degree of differential equation is 1 and 2 respectively.

80. Consider the differential equation,

$$\frac{dx}{dy} + 2yx = 2y$$

$$\frac{dx}{dy} = 2y(1 - x)$$

$$\int \frac{dx}{1 - x} = \int 2y dy$$

$$-\log(x - 1) = y^2 + c$$

The above equation passes through $(2, 0)$ so,

$$c = 0$$

So, the curve will be,

$$(x - 1) = e^{-y^2}$$

AP EAMCET-2018-Engineering-23 Apr-Shift-2

Solutions – Physics

81. The below table shows the significant figures of the given numbers.

Number	Significant Figures
74.083	5
0.029	2
0.002407	4
2.47×10^7	3

82. From the graph, centre of circle is (2, 0) and radius is 2.

The equation of the circle is,

$$(s - 2)^2 + (v - 0)^2 = 2^2$$

$$(s - 2)^2 + v^2 = 4$$

Differentiate above equation with respect to time,

$$2(s - 2)\frac{ds}{dt} + 2v\frac{dv}{dt} = 0$$

$$2(s - 2)v + 2va = 0$$

Apply the condition; at $s = 2 - \sqrt{2}$, $v = \sqrt{2}$

$$2(2 - \sqrt{2} - 2) \times \sqrt{2} + 2\sqrt{2}a = 0$$

$$a = \frac{4}{2\sqrt{2}}$$

$$a = \sqrt{2} \text{ ms}^{-2}$$

83. From the first condition,

$$30 = 10t_1$$

$$t_1 = 3 \text{ s}$$

From the second condition,

$$30t_2 = \frac{1}{2} \times 10t_2^2$$

$$t_2 = 6 \text{ s}$$

$$\text{So, } t_2 - t_1 = 3 \text{ s}$$

84. Initial velocity is calculated as,

$$v_i = u \cos \theta \hat{i} + u \sin \theta \hat{j}$$

$$v_i = 10\sqrt{3} \cos 60^\circ \hat{i} + 10\sqrt{3} \sin 60^\circ \hat{j}$$

$$v_i = 5\sqrt{3} \hat{i} + 15 \hat{j}$$

Final velocity is calculated as,

$$v_f = u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j}$$

$$v_f = 5\sqrt{3}\hat{i} - 5\hat{j}$$

The dot product of both velocities is,

$$v_i \cdot v_f = 25 \times 3 - 15 \times 5$$

$$v_i \cdot v_f = 0$$

The dot product is zero, so both the velocities are perpendicular to each other.

$$v_i \perp v_f$$

85. The expression of the normal force is,

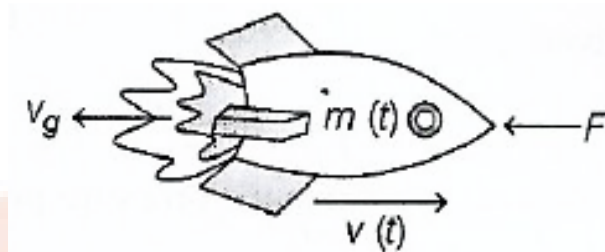
$$N = \sqrt{(k^2 x^2 + m^2 g^2)}$$

Substitute values in above expression.

$$N = \sqrt{\left(\left[\frac{(\sqrt{3} + 1)(0.100)(9.8)}{L} \right]^2 L^2 + (0.100)^2 (9.8)^2 \right)}$$

$$N = 2.55 \text{ N}$$

86. Consider the figure,



The motion equation for the rocket is,

$$m \frac{dv}{dt} = F_{ext} + v \frac{dm}{dt}$$

$$m \frac{dv}{dt} = v \frac{dm}{dt} \quad (\text{When } F_{ext} = 0)$$

Solve above equation,

$$\int \frac{dv}{v} = \int \frac{dm}{m}$$

$$\log v = \log m + C$$

Apply the condition, at $v = 0$, $m = m_0$

$$0 = \log m_0 + C$$

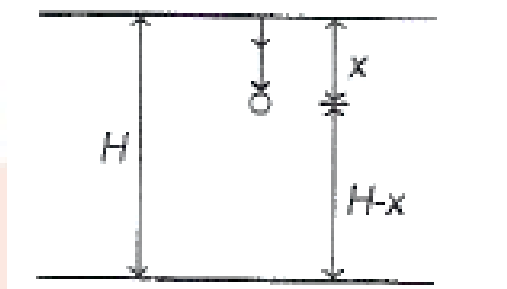
$$C = -\log m_0$$

Thus,

$$\frac{\Delta v}{v} = \log \frac{m}{m_0} dt$$

$$m = m_0 e^{-\frac{a}{v} t}$$

87. Consider the figure,



Given condition, $(KE) = 2(PE)$

Substitute values,

$$\frac{1}{2}mv^2 = 2mgh$$

$$\frac{1}{2}(\sqrt{2gx})^2 = 2g(H-x)$$

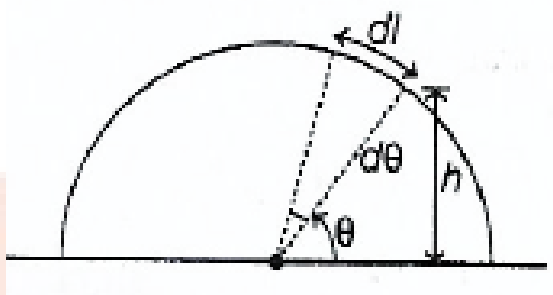
$$\frac{1}{2}(2gx) = 2g(H-x)$$

$$x = \frac{2H}{3}$$

Thus the height of the particle is $H - x = \frac{H}{3}$.

The speed of the particle is, $v = \sqrt{2g\left(\frac{2H}{3}\right)} = 2\sqrt{\frac{gH}{3}}$.

88. Consider the figure,



From figure,

$$\frac{h}{R} = \sin \theta \Rightarrow h = R \sin \theta$$

$$dl = R d\theta$$

The mass of dl length of the chain is, $dm = \frac{m}{l} dl$

The potential energy of the dm mass is,

$$dU = (dm)gh$$

$$dU = \frac{mgh}{l} dl$$

$$dU = \frac{mgh}{l} R d\theta$$

$$dU = \frac{mgR^2}{l} \sin \theta d\theta$$

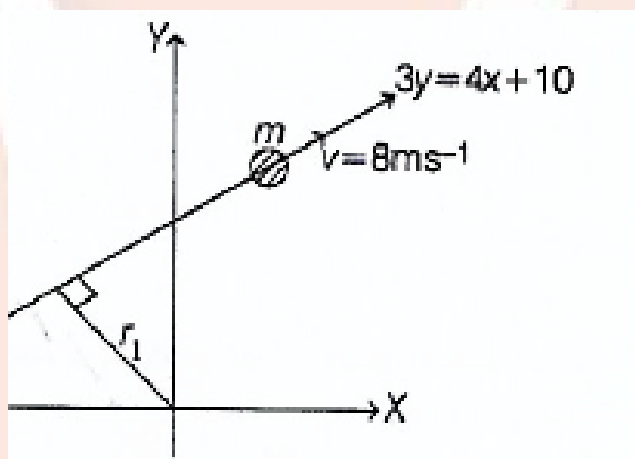
The potential energy of the complete chain is calculated as,

$$U = \int_{\pi/2}^{\pi/2-\theta} dU$$

$$U = \int_{\pi/2}^{\pi/2-\theta} \frac{mgR^2}{l} \sin \theta d\theta$$

$$U = \frac{mgR^2}{l} \sin\left(\frac{l}{R}\right)$$

89. Consider the figure,



The expression of the angular momentum is,

$$L = mvr_1$$

Substitute values.

$$L = 15 \times 8 \times \frac{10}{\sqrt{3^2 + 4^2}}$$

$$L = 240 \text{ kg} \cdot \text{m}^2 \text{s}^{-1}$$

90. The torque on the pulley is,

$$\tau = I\alpha$$

Substitute values.

$$mgR = \frac{MR^2\alpha}{2}$$

$$\alpha = \frac{2mg}{MR}$$

The acceleration of the rope is calculated as,

$$a = R\alpha$$

$$= R\left(\frac{2mg}{MR}\right)$$

$$= \frac{2mg}{M}$$

Substitute values.

$$a = \frac{2(1)(10)}{8}$$

$$a = \frac{10}{4} \text{ ms}^{-2}$$

The speed of bucket is calculated as,

$$v = \sqrt{2aS}$$

$$v = \sqrt{2\left(\frac{10}{4}\right)(16)}$$

$$v = 8.6 \text{ ms}^{-1}$$

$$v \approx 8 \text{ ms}^{-1}$$

91. Given, $y = 5 \sin\left(4t + \frac{\pi}{3}\right)$

So, the angular velocity is, $\omega = 4$

The time period is, $T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = \frac{\pi}{2} \text{ s}$

The time is, $t = \frac{T}{4} = \frac{\pi}{8} \text{ s}$

The expression of velocity is,

$$v = \frac{dy}{dt}$$

$$v = \frac{d}{dt} \left[5 \sin\left(4t + \frac{\pi}{3}\right) \right]$$

$$v = 20 \cos\left(4t + \frac{\pi}{3}\right)$$

The velocity at $t = \frac{\pi}{8} \text{ s}$ is,

$$v = 20 \cos \left(4 \left(\frac{\pi}{8} \right) + \frac{\pi}{3} \right)$$

$$v = -\frac{20\sqrt{3}}{2} \text{ ms}^{-1}$$

The kinetic energy of particle is calculated as,

$$K = \frac{1}{2}mv^2$$

$$K = \frac{1}{2} \left(\frac{2}{1000} \right) \left(-\frac{20\sqrt{3}}{2} \right)^2$$

$$K = 0.3 \text{ J}$$

92. The expression of the initial force is,

$$F_1 = \frac{Gm^2}{r^2}$$

The masses after the transformation of 20% mass from one body to another are,

$$m_1 = \frac{80}{100}m \quad \text{and} \quad m_2 = \frac{120}{100}m$$

The expression of final force is,

$$F_2 = \frac{Gm^2}{r^2} \left(\frac{24}{25} \right)$$

The reduction in final force is calculated as,

$$\begin{aligned}
 \% \text{reduction} &= \frac{F_1 - F_2}{F_1} \times 100\% \\
 &= \left(1 - \frac{F_2}{F_1} \right) \times 100\% \\
 &= -4\%
 \end{aligned}$$

93. The spring constant for rod is,

$$k_1 = \frac{YA}{L}$$

The equivalent spring constant after series connection of rod and spring is calculated as,

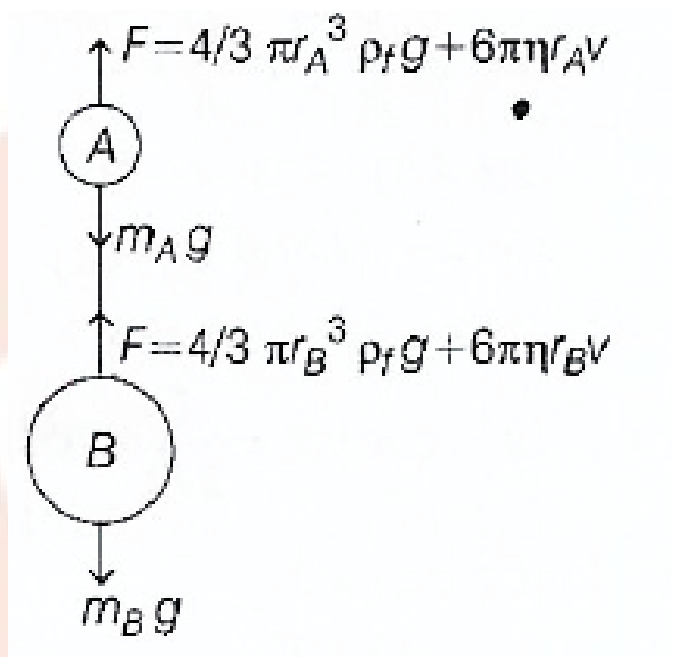
$$\begin{aligned}
 k_{eq} &= \frac{k_1 k_2}{k_1 + k_2} \\
 &= \frac{kYA / L}{k + \frac{YA}{L}} \\
 &= \frac{kYA}{kL + YA}
 \end{aligned}$$

The time period of oscillation is,

$$T = 2\pi \sqrt{\frac{m}{k_{eq}}}$$

$$T = 2\pi \sqrt{\frac{m(kL + YA)}{kYA}}$$

94. The free body diagram of the system is shown below.



The condition when the system achieves terminal velocity is,

$$F_{net} = 0$$

$$\frac{4}{3} \pi \rho_f g (r_A^3 + r_B^3) + 6\pi\eta v (r_A + r_B) = (m_A + m_B) g$$

Substitute values.

$$\left[\frac{4}{3} \pi (2800)(10) \left[\begin{array}{l} (0.002)^3 \\ + (0.004)^3 \end{array} \right] + 6\pi(1)v(0.002 + 0.004) \right] = (2800) \frac{4}{3} \pi \left[\begin{array}{l} (0.002)^3 \\ + (0.004)^3 \end{array} \right] (10)$$

$$v = 0.04 \text{ ms}^{-1}$$

$$v = 4 \text{ cms}^{-1}$$

95. A refrigerator (works as a heat engine) is a device that extracts heat from a low temperature reservoir and transfers it to a high temperature reservoir.

So, after opening of the door of refrigerator, it exhausts more heat into the room than it extracts from it. Hence, the room gets hotter.

Thus, the assertion is false but reason is true.

96. From the conservation of energy,

Heat Gain = Heat Generated

$$mL = I^2 Rt$$

$$m = \frac{I^2 Rt}{L}$$

Substitute values.

$$m = \frac{(10)^2 (20)(60)}{79.7 \times 4.2}$$

$$m = 359 \text{ g}$$

97. From the graph,

$$V \propto T$$

$$\frac{V}{T} = \text{constant}$$

Thus, the process is a constant pressure process.

The expression of the work done is,

$$\Delta W = p\Delta V$$

$$\Delta W = nR\Delta T$$

$$\Delta W = nR \left(\frac{\Delta Q}{nC_p} \right)$$

$$\Delta W = \frac{\Delta Q R}{\frac{5}{2}R} = \frac{2}{5} \Delta Q$$

Substitute values,

$$\Delta W = \frac{2}{5} (5 \text{ cal})$$

$$\Delta W = (2 \text{ cal}) \left(\frac{4.2 \text{ J}}{1 \text{ cal}} \right)$$

$$\Delta W = 8.4 \text{ J}$$

98. Given equation, $pV^{\frac{3}{2}} = \text{constant} \Rightarrow \gamma = \frac{3}{2}$

From the gas equation,

$$TV^{\gamma-1} = \text{constant}$$

$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{\gamma-1}$$

Substitute values.

$$T_f = T \left(\frac{2V_f}{V_f} \right)^{\frac{3}{2}-1}$$

$$T_f = \sqrt{2}T$$

99. The expression of the rms velocity is,

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

For temperature T_1 , the rms velocity is,

$$600 = \sqrt{\frac{3RT_1}{M}} \quad \text{..... (I)}$$

For temperature T_2 , the rms velocity is,

$$v'_{rms} = \sqrt{\frac{3R(2T_1)}{(M/2)}}$$

$$v'_{rms} = 2\sqrt{\frac{3RT_1}{M}} \quad \text{..... (II)}$$

$$v'_{rms} = 2v_{rms}$$

$$v'_{rms} = 2(600)$$

$$v'_{rms} = 1200 \text{ ms}^{-1}$$

100. The wavelength is calculated as,

$$\lambda = \frac{v}{f}$$

$$\lambda = \frac{360}{500}$$

$$\lambda = \frac{36}{50}$$

The path difference is,

$$\Delta x = \frac{\theta}{360^\circ} \times \lambda$$

Substitute values.

$$\Delta x = \frac{60^\circ}{360^\circ} \times \left(\frac{36}{50} \right)$$

$$\Delta x = 0.12 \text{ m}$$

101. The maximum speed of O is,

$$\begin{aligned}v_m &= a\omega \\&= (0.5)(40) \\&= 20 \text{ ms}^{-1}\end{aligned}$$

The expression of the maximum frequency is,

$$f_{\max} = f \left(\frac{c + v_{\text{observer}}}{c - v_{\text{source}}} \right)$$

Substitute values,

$$\begin{aligned}f_{\max} &= (288) \left(\frac{340 + 20}{340 - 20} \right) \\f_{\max} &= 324 \text{ Hz}\end{aligned}$$

102. Given, for object, $u = -2 \text{ cm}$ and $f = 1.5 \text{ cm}$

The lens formula is,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Substitute values.

$$\frac{1}{v} - \frac{1}{(-2)} = \frac{1}{1.5}$$

$$v = 6 \text{ cm}$$

Given, for eyepiece,

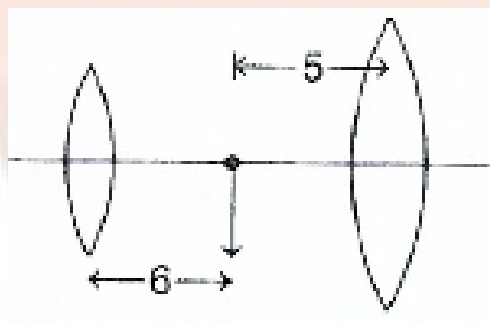
$$v = -25 \text{ cm and } f = 6.25 \text{ cm}$$

By the lens formula,

$$\frac{1}{u} = \frac{1}{-25} - \frac{1}{6.25}$$

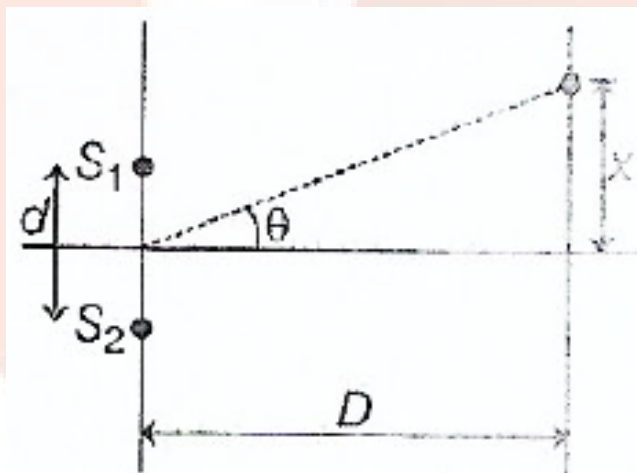
$$u = -5 \text{ cm}$$

The distances is shown below,



Thus the distance between two lenses is 11 cm.

103. Consider the figure,



The expression of the intensity is,

$$I = 4I_0 \cos^2 \frac{\phi}{2} \quad \dots\dots (I)$$

Here,

$$\phi = \frac{2\pi d}{\lambda} \sin \theta$$

$$\phi = \frac{2\pi d}{\lambda} \left(\frac{x}{D} \right)$$

$$\phi = \frac{2\pi x}{\left(\frac{\lambda D}{d} \right)}$$

$$\phi = \frac{2\pi x}{\beta}$$

From equation (I),

$$I = 4I_0 \cos^2 \left(\frac{\pi x}{\beta} \right)$$

104. The acceleration of α particle is,

$$a_r = \frac{qE}{m} = \frac{2eE}{4}$$

The acceleration of photon is,

$$a_p = \frac{eE}{1}$$

$$a_p = eE$$

Both the particles travel same distance in time t_1 and t_2 . So, from the displacement equation,

$$\frac{1}{2} \left(\frac{2eE}{4} \right) t_1^2 = \frac{1}{2} \left(\frac{eE}{1} \right) t_2^2$$

$$\frac{t_2^2}{t_1^2} = \frac{1}{2}$$

$$\frac{t_2}{t_1} = \frac{1}{\sqrt{2}}$$

105. In the question, the information about the masses of balls is not given.

106. Given,

$$\vec{E} = (3\hat{i} + 4y\hat{j}) \text{ Vm}^{-1} \quad \text{..... (I)}$$

The standard equation of electric field is,

$$\vec{E} = \frac{-\partial V_x}{\partial x} \hat{i} + \frac{-\partial V_y}{\partial y} \hat{j} \quad \text{..... (II)}$$

From equation (I) and (II),

$$\frac{-\partial V_x}{\partial x} = 3 \Rightarrow V_x = -3x$$

$$\frac{-\partial V_y}{\partial y} = 4y \Rightarrow V_y = -2y^2$$

The potential function is,

$$V = -(3x + 2y^2)$$

The potential function at point (2, 1) is,

$$V = -(3 \times 2 + 2 \times 1^2)$$

$$V = -8 \text{ V}$$

107. The equivalent capacitance of the system is calculated as,

$$\frac{1}{C_{eq}} = \frac{1}{3} + \frac{1}{4+2}$$

$$\frac{1}{C_{eq}} = \frac{1}{3} + \frac{1}{6}$$

$$C_{eq} = 2 \mu\text{F}$$

The charge taken from source is,

$$q_{eq} = C_{eq} \Delta V$$

$$q_{eq} = (2 \times 10^{-6})(1800)$$

$$q_{eq} = 3600 \mu\text{C}$$

The potential drop across capacitor C_1 is,

$$C_1 = \frac{q_{C_1}}{C_{C_1}}$$

$$C_1 = \frac{3600 \times 10^{-6}}{3 \times 10^{-6}}$$

$$C_1 = 1200 \text{ V}$$

The potential drop across combination of $4 \mu\text{F}$ and $2 \mu\text{F}$ capacitors is,

$$\Delta V' = 1800 - 1200$$

$$\Delta V' = 600 \text{ V}$$

The charge on $4 \mu\text{F}$ capacitor is,

$$q_2 = C_2 \Delta V'$$

$$q_2 = (4 \times 10^{-6})(600)$$

$$q_2 = 2.4 \times 10^{-3} \text{ C}$$

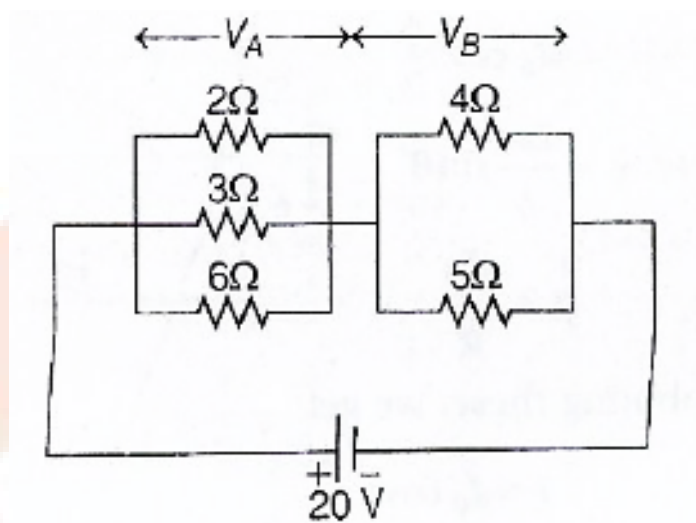
The charge on $2 \mu\text{F}$ capacitor is,

$$q_3 = C_3 \Delta V'$$

$$q_3 = (2 \times 10^{-6})(600)$$

$$q_3 = 1.2 \times 10^{-3} \text{ C}$$

108. Consider the circuit diagram,



As, the resistance of portion B is higher, so there is greater potential drop across B occur.

The expression of the power output is,

$$P = \frac{V^2}{R}$$

From the above equation, least value of the resistance results in higher power output.

Thus, $4\ \Omega$ bulb will glow with maximum intensity.

109. The distribution of current in circuit is shown below,



The expression of power across the resistors is,

$$P_1 = I_1^2 R_1 = \frac{4}{9} i^2$$

$$P_2 = I_2^2 R_2 = \frac{2}{9} i^2$$

$$P_3 = I_3^2 R_3 = 3i^2 = \frac{27}{9} i^2$$

The ratio of power is,

$$P_1 : P_2 : P_3 = 4 : 2 : 27$$

110. The expression of force is,

$$F = \frac{\mu_0 i_1 i_2}{2\pi x}$$

The expression of the work done is,

$$W = \int_d^{2d} F dx$$

Substitute values.

$$W = \int_d^{2d} \frac{\mu_0 i_1 i_2}{2\pi x} dx$$

$$W = \frac{\mu_0 i_1 i_2}{2\pi} [\log_e x]_d^{2d}$$

$$W = \frac{\mu_0 i_1 i_2}{2\pi} (\log_e 2)$$

111. The expression of the magnetic induction is,

$$B = \frac{\mu_0 I}{4\pi r} (\sin \theta_1 + \sin \theta_2)$$

Substitute values,

$$B = \frac{(4\pi \times 10^{-7})(30)}{4\pi(12 \times 10^{-2})} \left(\frac{16}{20} + \frac{16}{20} \right)$$

$$B = 4 \times 10^{-5} \text{ T}$$

$$B = 0.4 \text{ gauss}$$

112. The total dipole moment of sample is calculated as,

$$\mu = N \mu_0 (c)$$

$$\mu = (3 \times 10^{24}) \times (2 \times 10^{-23}) \times \left(\frac{10}{100} \right)$$

$$\mu = 6 \text{ JT}^{-1}$$

From Curie's law,

$$m \propto \frac{B}{T}$$

$$\frac{B_1}{T_1} = \frac{B_2}{T_2}$$

Substitute values.

$$\frac{6 \times \frac{990}{880}}{2.1} = \frac{B_2}{3.5}$$

$$B_2 = 11.25 \text{ A} \cdot \text{m}^2$$

113. The expression of the self inductance of coil is,

$$L = \frac{N\phi_B}{I}$$

$$L = \frac{N}{I} \frac{\mu_0 I}{2R} \times \pi R^2$$

$$L = \frac{\pi \mu_0 N R}{2}$$

$$L \propto R$$

Thus,

$$\frac{L_2}{L_1} = \frac{N_2}{N_1}$$

Substitute values,

$$\frac{L_2}{108} = \frac{500}{600}$$
$$L_2 = 90 \text{ mH}$$

114. From the given expression of emf and current, the current leads by $\frac{\pi}{4}$. So, the circuit must be maximum capacitive than inductive. Thus, the circuit is either C, R series combination or L, C and R series combination.

115. The wave number from the equation of wave is,

$$k = 300\pi$$

Substitute the formula of wave number,

$$\frac{2\pi}{\lambda} = 300\pi$$

$$\lambda = 6.67 \times 10^{-3} \text{ m}$$

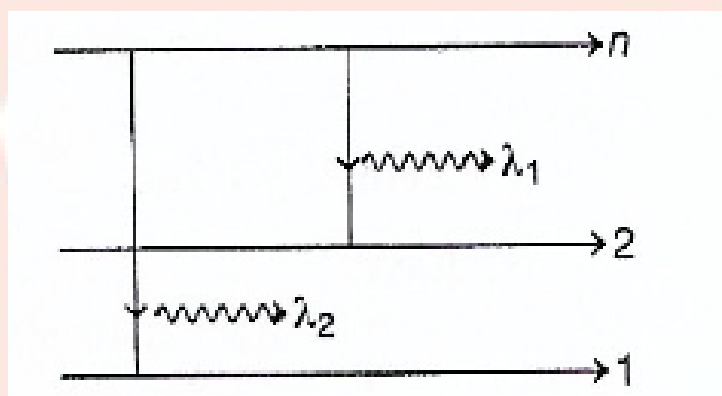
116. In the process of photoemission, one electron produces only one proton.

The percentage of incident photons which produces current in photocells is,

$$I = \frac{n}{t}$$

$$I = \frac{100Ihc}{eP\lambda}$$

117. Consider the figure,



The expression of wavelength λ_1 is,

$$\lambda_1 = \frac{hc}{\Delta E_{n \rightarrow 2}} \quad \text{..... (I)}$$

The expression of wavelength λ_2 is,

$$\lambda_2 = \frac{hc}{\Delta E_{2 \rightarrow 1}} \quad \text{..... (II)}$$

From equation (I) and (II),

$$\lambda_1 \left(\frac{13.6}{4} - \frac{13.6}{n^2} \right) = \lambda_2 \left(\frac{13.6}{1} - \frac{13.6}{n^2} \right)$$

$$n = \sqrt{\frac{4(\lambda_2 - \lambda_1)}{4\lambda_2 - \lambda_1}}$$

118. From the expression of the radioactive decay,

$$N = N_0 \left(\frac{1}{2} \right)^n$$

Substitute values.

$$\frac{(100 - 93.75)N_0}{100} = N_0 \left(\frac{1}{2} \right)^n$$

$$\frac{6.25N_0}{100} = N_0 \left(\frac{1}{2} \right)^n$$

$$\left(\frac{1}{2} \right)^n = \frac{625}{10000}$$

$$n = 4$$

119. Given, the current amplification factor is, $\alpha = 0.95$

The gain is calculated as,

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\beta = \frac{0.95}{1 - 0.95}$$

$$\beta = 19$$

The change in collector current is calculated as,

$$\Delta I_C = \beta \Delta I_B$$

$$\Delta I_C = (19)(2)$$

$$\Delta I_C = 38 \mu\text{A}$$

120. The expression of the area covered by transmission is,

$$A = 2\pi Rh$$

$$A \propto h$$

Thus, if the height of tower is increased by 30%, the area covered by transmission is also increased by 30%.

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Solutions – Chemistry

121. The total energy can be calculated as:

$$\begin{aligned}\text{Total energy} &= \omega_o + \text{KE} \\ &= 4.80 + 0.220 \\ &= 5.0 \text{ eV}\end{aligned}$$

Since we know that 1 eV is equal to $1.6021 \times 10^{-19} \text{ J}$

Therefore, 5 eV will be equal to

$$5 \times 1.6021 \times 10^{-19} \text{ J} = 8.010 \times 10^{-19} \text{ J}$$

The frequency is calculated as:

$$\text{Frequency}(\nu) = \frac{\text{Total Energy}}{h(\text{Plancks constant})}$$

Substitute the values

$$\begin{aligned}\text{Frequency}(\nu) &= \frac{8.010 \times 10^{-19}}{6.626 \times 10^{-34}} \\ &= 1.21 \times 10^{15} \text{ Hz}\end{aligned}$$

122. The ratio can be calculated as:

$$\frac{r_{(\text{H})}}{r_{(\text{Li}^{2+})}} = \frac{Z_{(\text{Li}^{2+})}}{Z_{(\text{H})}}$$
$$= \frac{3}{1}$$

123. (i) In the periodic table, around 78% of elements are metals.

(ii) The metallic character increases as we move from top to bottom in a group and non-metallic character increases as we move from left to right in a period.

(iii) The element Ho belongs to f-block.

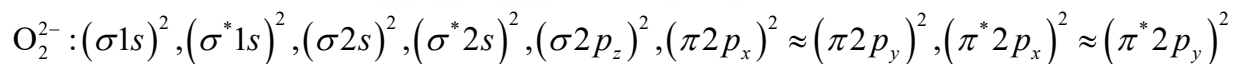
Therefore, statements (i) and (iii) are correct.

124. The dipole moment of H_2O is higher than NH_3 and NF_3 because oxygen being more electronegative than nitrogen withdraws the bonding electrons to a greater extent.

Therefore, the correct order is,



125. The molecular orbital configuration of species O_2^{2-} is shown below.



In bonding orbitals, there are total 10 electrons.

In antibonding orbitals, there are total 8 electrons.

126. Since we know that the most probable speed of a compound is inversely proportional to the square root of its molar mass.

Therefore,

$$M_2 > M_1 > M_3$$

127. Molarity of HCl (M_1) is 0.1 M

Molarity of Na_2CO_3 (M_2) can be calculated as:

$$\begin{aligned} M_2 &= \frac{\text{Mass of } Na_2CO_3 \times 1000}{\text{Molar mass} \times \text{Volume (L)}} \\ &= \frac{0.106 \times 1000}{106 \times 20} \\ &= 0.05 \end{aligned}$$

The volume of HCl (V_1) required can be calculated as:

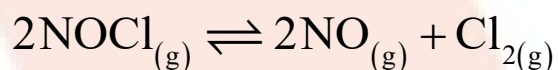
$$\begin{aligned}M_1V_1Z_1 &= M_2V_2Z_2 \\V_1 &= \frac{M_2V_2Z_2}{M_1Z_1} \\&= \frac{0.05 \times 20 \times 2}{0.1 \times 1} \\&= 20 \text{ mL}\end{aligned}$$

128. The molar mass of carbon dioxide is 44 g/mol.

Therefore, heat released by CO_2 is:

$$\begin{aligned}\text{Heat released} &= \frac{-394 \times 17.6}{44} \\&= -157.6 \text{ kJ}\end{aligned}$$

129. The given reaction is:



For the above reaction, the value of Δn can be calculated as:

$$\begin{aligned}\Delta n &= \text{Moles of gaseous products} - \text{Moles of gaseous reactants} \\&= 3 - 2 \\&= 1\end{aligned}$$

The equilibrium constant (K_c) can be calculated as:

$$K_p = K_c (RT)^{\Delta n}$$

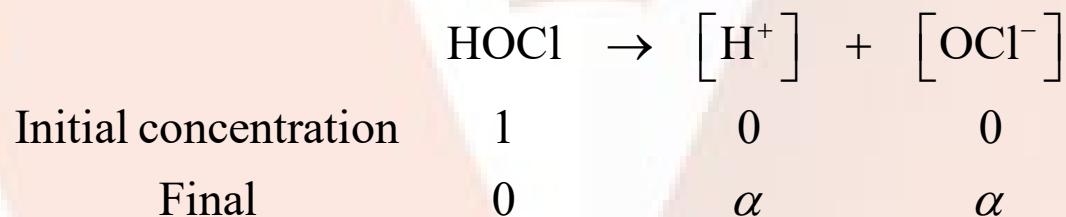
$$4.157 \times 10^{-4} = K_c (0.083 \times 1000)^1$$

$$K_c = 5 \times 10^{-6}$$

130. First, we will calculate degree of ionization as follows:

$$\begin{aligned}\alpha &= \sqrt{\frac{K_a}{C}} \\ &= \sqrt{\frac{2.5 \times 10^{-5}}{1}} \\ &= 5 \times 10^{-3}\end{aligned}$$

The dissociation of HCl is shown below:

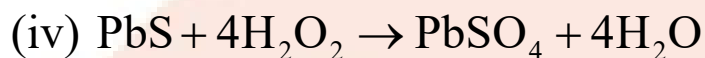
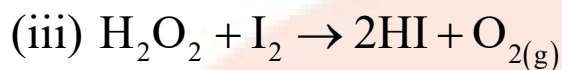
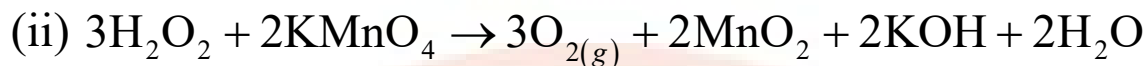
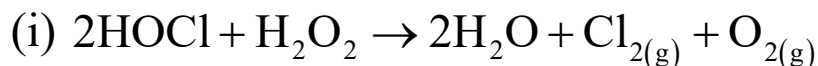


The concentration of H^+ ions is equal to $\alpha = 5 \times 10^{-3}$.

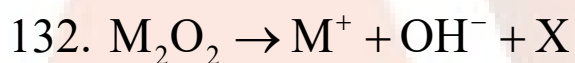
The pH of solution can be calculated as :

$$\begin{aligned}\text{pH} &= -\log[\text{H}^+] \\ &= -\log(5 \times 10^{-3}) \\ &= 2.3\end{aligned}$$

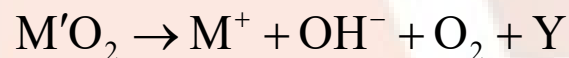
131. The reactions are shown below.



Therefore, in reaction (iv), oxygen gas does not liberate.



In the above reaction, X is O_2



In the above reaction, Y is H_2O_2

133. (i) The atomic radius of Al is higher than the atomic radius of Ga. It is due to the presence of greater screening effect in aluminium. Ga shows poor screening effect.

(ii) Many allotropic forms are shown by boron such as α – rhombohedral, β – rhombohedral and β – tetragonal.

(iii) As we move from top to bottom in a group, melting point decreases. Therefore, Ga shows lowest melting point in group.

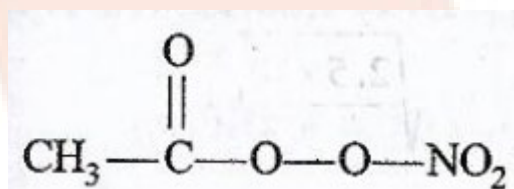
Therefore, statements (ii) and (iii) are correct.

134. GeX_2 is not more stable than GeX_4 .

GeX_4 shows greater screening effect.

Therefore, statement (2) is not correct.

135. Peroxyacetyl nitrate is a compound which is mainly responsible for causing eye irritation in photochemical smog. The structure of peroxyacetyl nitrate is shown below.

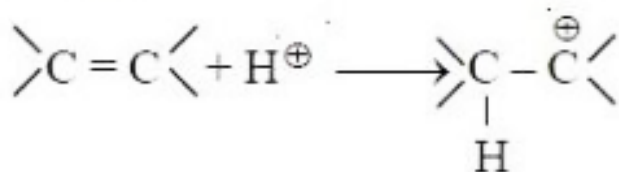


136. A) The resonance effect is shown by benzene (C_6H_6)

B) Inductive effect is shown by 1-chloropropane



C) Electromeric effect is shown by ethene as shown below.



D) Hyperconjugation effect is shown by ethane as shown below.



Therefore,

A	B	C	D
III	V	I	II

137. In dehydrohalogenation reaction, the order of reactivity order is
Tertiary > Secondary > Primary

The reactivity order of alkyl halide is



Therefore, the rate of dehydrohalogenation of $CH_3CH_2CH_2Cl$ will be less.

138. When propyne reacts with HBr, it forms 2,2 dibromo propane as shown below,.

$$x = \frac{334000}{1.67 \times 10^8}$$
$$= 2 \times 10^{-3}$$

The quantity of CO_2 can be derived as follows:

$$\begin{aligned}\text{Mole fraction}(x) &= \frac{\frac{m_{\text{CO}_2}}{M_{\text{CO}_2}}}{\frac{m_{\text{H}_2\text{O}}}{M_{\text{H}_2\text{O}}}} \\ 2 \times 10^{-3} &= \frac{m_{\text{CO}_2} \times 18}{44 \times 500} \\ m_{\text{CO}_2} &= \frac{2 \times 10^{-3} \times 44 \times 500}{18} \\ &= 2.442 \text{ g}\end{aligned}$$

141. The osmotic pressure is calculated as:

$$\pi = iCRT$$

$$\pi = \frac{i \times m \times RT}{M \times V(\text{in mL})} \times 1000$$

Substitute all the values in the above expression.

$$\begin{aligned}5.04 \times 10^{-3} &= \frac{1 \times 2.52 \times 0.08206 \times 300}{M \times 300} \times 1000 \\ M &= \frac{1 \times 2.52 \times 0.08206 \times 300 \times 1000}{300 \times 5.04 \times 10^{-3}} \\ &= \frac{62037.36}{1.512} \\ &= 41 \times 10^3\end{aligned}$$

142. First we will calculate conductivity as follows:

$$\begin{aligned}\text{Conductivity} &= 3.12 \times 10^{-4} \times 0.5 \text{ S cm}^{-1} \\ &= 1.56 \times 10^{-4} \text{ S cm}^{-1}\end{aligned}$$

The molar conductivity becomes equal to

$$\begin{aligned}\text{Molar conductivity} &= \frac{1.56 \times 10^{-4}}{0.01} \\ &= 1.56 \text{ S cm}^3 \text{ mol}^{-1}\end{aligned}$$

$$\begin{aligned}\text{Degree of dissociation} &= \frac{1.56}{390} \\ &= 0.004\end{aligned}$$

Now, dissociation constant is calculated as:

$$\begin{aligned}\text{Dissociation constant} &= \frac{(0.004)^2}{1 - (0.004)^2} \\ &= 1.6 \times 10^{-5}\end{aligned}$$

143. First we will compare expt. (1) and expt. (2)

The rate of reaction gets double when [B] is doubled

Therefore, reaction is of first order in B.

Now, we will compare expt. (2) and expt. (3)

The rate of reaction gets double when [C] is doubled

Therefore, reaction is of first order in C.

Now, compare expt. (3) and expt. (4)

The rate of reaction does not change when [A] is doubled

Therefore, reaction is of zero order in A.

The rate law expression can be written as:

$$\text{Rate law} = k[A]^0[B]^1[C]^1$$

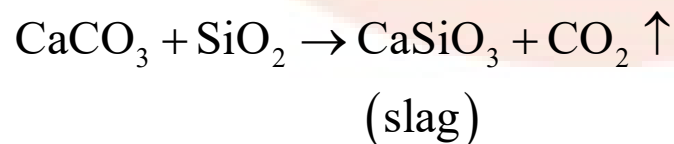
By inserting data of expt. (1) in the above expression, we obtain

$$2.4 \times 10^{-6} = k[0.02]^0[0.1]^1[0.003]^1$$

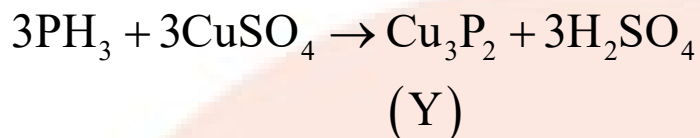
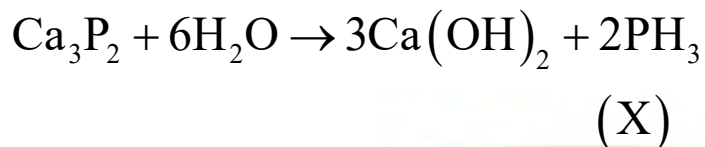
$$\begin{aligned} k &= \frac{2.4 \times 10^{-6}}{0.1 \times 0.03} \\ &= 8 \times 10^{-4} \text{ L mol}^{-1} \text{ s}^{-1} \end{aligned}$$

144. In the hydrogenation of vegetable oil, the reactants and catalyst exist in three different states.

145. Iron ores consists acidic impurities such as silica which react with calcium carbonate to produce molten slag as shown below.



146. The reactions taking place are shown below.



147. (i) ZnO , PbO and Sb_2O_3 show characteristics of both acids and bases as well. Therefore, they all amphoteric oxides.

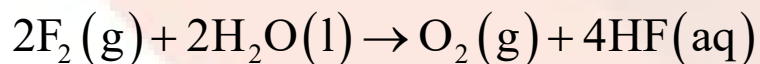
(ii) CO and NO are oxygen poor compounds of non-metals.

Therefore, they all are neutral oxides.

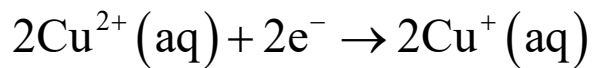
(iii) CrO_3 , Mn_2O_7 , V_2O_5 form acidic oxides.

Thus, all statements are incorrect.

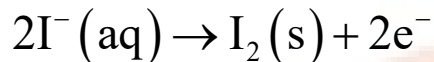
148. When fluoride reacts with water, it liberates oxygen as shown below.



149. The negative cell potential (non-spontaneous) is shown by the reaction of Cu^{2+} with I^- . For the reaction, the half-cell reduction reaction is:



The standard potential for the reaction is +0.154 V

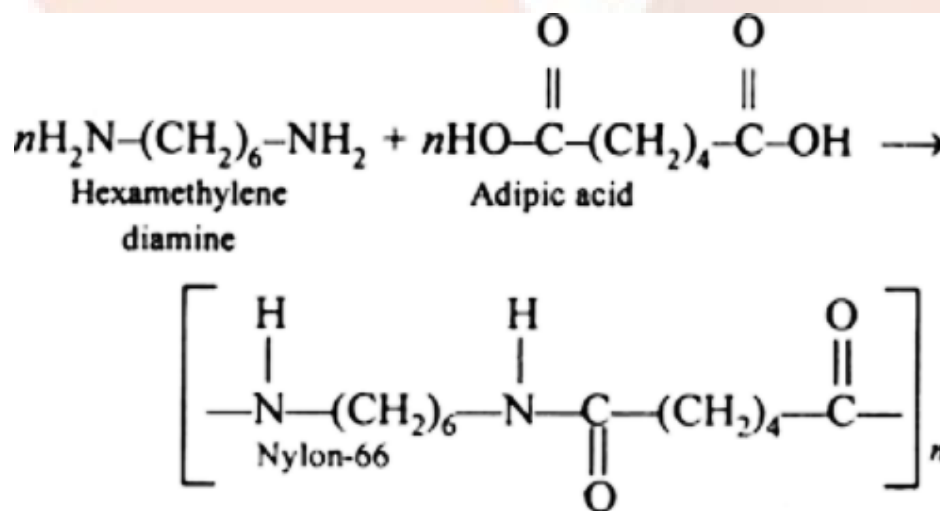


The standard potential for the reaction is -0.535 V

Therefore, the reaction is not possible.

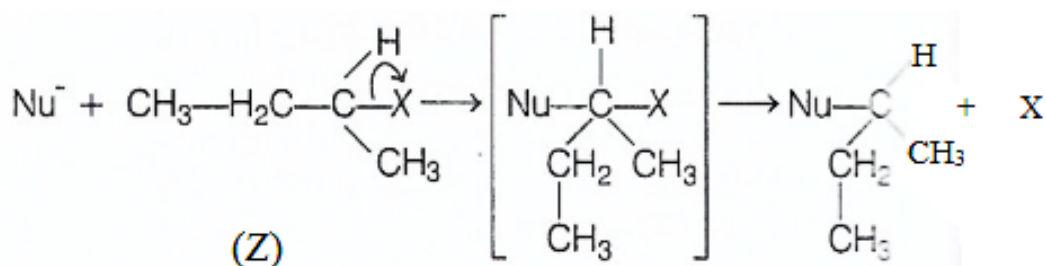
150. No precipitate of AgCl is attained on reacting the compound with AgNO_3 . Therefore, the correct formula is $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ which is having coordination number 6, therefore, the secondary valency is 6 as well.

151. The condensation reaction between nylon 6,6 produces Nylon-6,6 as shown below.

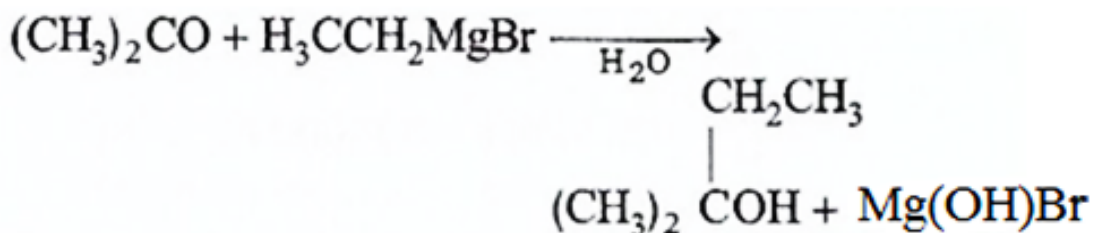


There are total six CH_2 groups in hexamethylene diamine and four CH_2 groups in adipic acid.

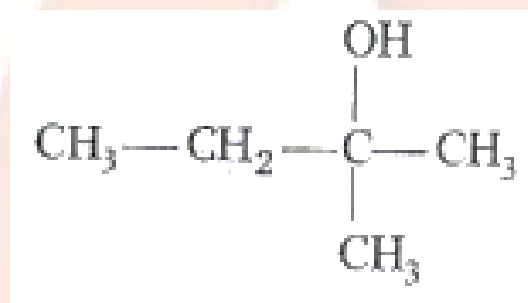
152. Among the given saccharides, the reducing saccharides are ribose, maltose and lactose as they are having free aldehydic or ketonic groups.
153. Examples of antihistamine and cationic detergent are dimethane and cetyl trimethyl ammonium chloride respectively.
154. In $\text{S}_\text{N}2$ reaction, the nucleophile attacks the electrophilic carbon of the electrophile. An intermediate is formed and as a result, the bond between carbon and leaving group breaks as shown below.



155. When ethyl magnesium bromide reacts with acetone, it forms 2-methyl butan-2-ol as shown below.

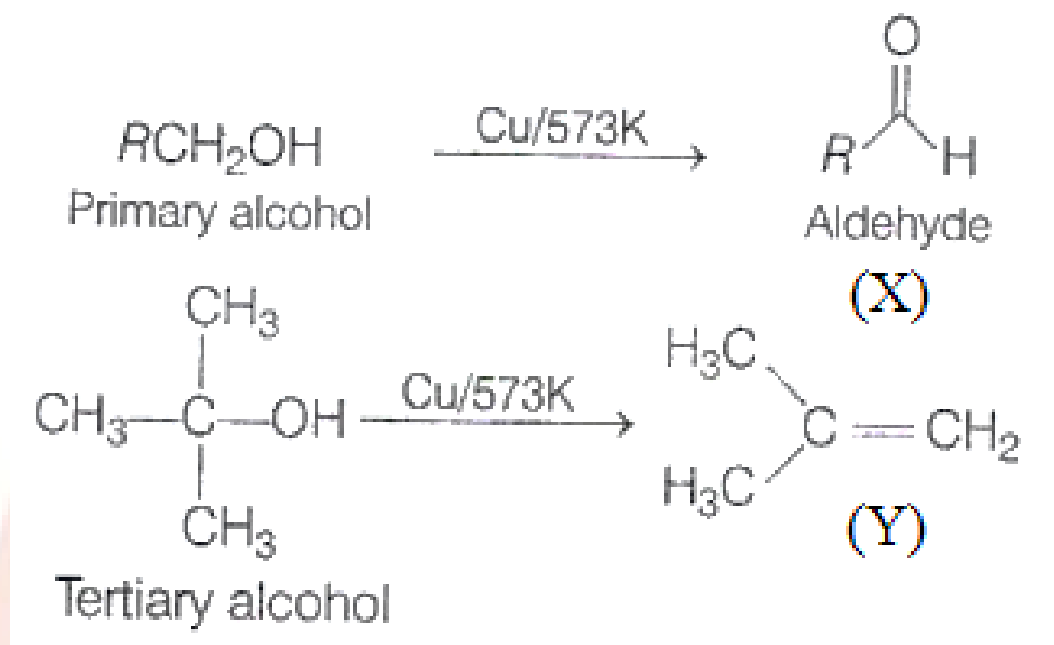


The structure of 2-methyl butan-2-ol is as follows:

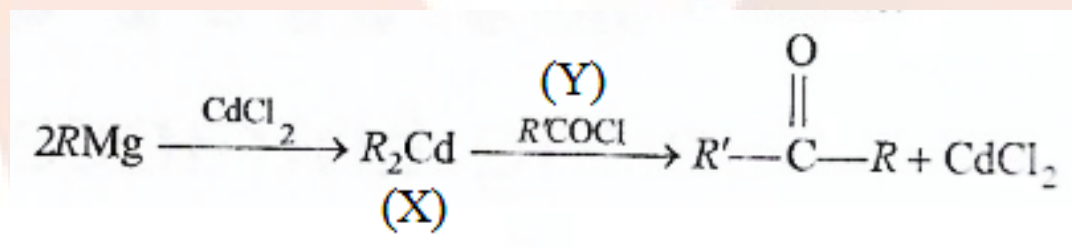


156. In o-nitrophenol, intramolecular bonding is available as nitro group is close to hydroxyl group. Therefore, the abstraction of proton becomes difficult. Hence, its pK_a will be least.

157. The products (X) and (Y) formed are shown below.



158. The products (X) and (Y) formed are shown below.



159. As we know that the electron releasing groups decreases the strength of acids and electron withdrawing groups increases the strength of acids.

Nitro group ($-\text{NO}_2$) is an electron withdrawing group.

Methoxy ($-\text{OCH}_3$) is an electron releasing group.

Acetic acid (CH_3COOH) is an electron releasing group.

Therefore, the correct order is:

IV > I > II > III

160. The products (X), (Y) and (Z) formed are shown below:

